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Single-mRNA imaging and modeling reveal coupled translation initiation and elongation rates

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eLife Assessment

This **important** study provides evidence for dynamic coupling between translation initiation and elongation that can help maintain low ribosome density and translational homeostasis. The authors combine single-molecule imaging with a new approach to analyze mRNA translation kinetics using Bayesian modeling. This work is overall **solid** and will be of interest to those studying translational regulation.

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Abstract

mRNA translation involves multiple regulatory steps, but how translation elongation influences protein output remains unclear. Using SunTag live-cell imaging and mathematical modeling, we quantified translation dynamics in single mRNAs across diverse coding sequences. Our Totally Asymmetric Exclusion Process (TASEP)-based Hidden Markov Model (HMM) revealed a strong coordination between initiation and elongation rates, resulting in consistently low ribosome density (≤12% occupancy) across all reporters. This coupling persisted under pharmacological inhibition of the elongation factor eIF5A, where proportional decreases in both initiation and elongation rates maintained homeostatic ribosome density. In contrast, eIF5A knockout cells exhibited a significant decrease in ribosome density, suggesting altered coordination. Together, these results highlight a dynamical coupling of initiation and elongation rates at the single-mRNA level, preventing ribosome crowding and maintaining translational homeostasis in mammalian cells.

Introduction

Regulation of protein synthesis is essential for cellular homeostasis and adaptive responses to external stimuli. While the untranslated regions (UTRs) of messenger RNA (mRNA) have traditionally been recognized for their central role in translation initiation (Hinnebusch et al., 2016; Mayr, 2018), the influence of the coding sequence (CDS) on initiation remains unclear. Indeed, in a simple model where elongation is much faster than initiation, heterogeneity in elongation rates has no impact on the total protein output per mRNA (Plotkin and Kudla, 2011; Shah et al., 2013; Erdmann-Pham et al., 2020). This view is supported by measurements indicating that initiation happens a few times per minute, while elongation proceeds at 2–6 aa/s (Boersma et al., 2019; Yan et al., 2016; Mateju et al., 2020; Livingston et al., 2023; Madern et al., 2025), suggesting that initiation is the rate-limiting step in mammalian translation.

However, this paradigm is being challenged by recent findings highlighting the CDS itself as a critical regulator of translational dynamics. In particular, specific CDS features can affect ribosome elongation, triggering surveillance pathways that sense ribosome stalling and collisions (Juszkiewicz et al., 2020; Li et al., 2022; Barrington et al., 2023; Bicknell et al., 2024;

Madern et al., 2025). Some of these pathways, such as ribosome quality control (RQC), can induce ribosome recycling and mRNA degradation (Wu et al., 2020; Goldman et al., 2021; Buschauer et al., 2020), whereas others down-regulate translation initiation in *cis* by recruiting regulatory complexes to stalled ribosomes (Amaya Ramirez et al., 2018; Hickey et al., 2020; Juszkiewicz et al., 2020). This suggests that cells have the capacity to monitor elongation rates and ribosome density at the level of individual mRNAs, responding by adjusting initiation or degrading aberrant transcripts (Joazeiro, 2019). Nevertheless, a comprehensive quantitative understanding of how elongation rates feedback on initiation in mammalian systems, at both bulk and single-mRNA levels, is still lacking. Furthermore, it is still unclear whether this coupling mechanism allows cells to dynamically adapt translation in response to environmental or physiological stress, such as nutrient deprivation, oxidative stress, or inhibition of specific translation factors. Clarifying this would shed light on the broader role of initiation–elongation feedback in translational homeostasis.

Elongation rates along transcripts depend on several factors including aminoacyl-tRNA availability, codon-anticodon interactions, co-translational folding, and biochemical properties of certain amino acids (Neelagandan et al., 2020; Madern et al., 2025). Notably, peptide-bond formation is particularly slow with multiple proline (P) residues, as proline is both a weak donor and acceptor in peptidyl transfer (Lassak et al., 2015). With poly-proline motifs being abundant in mammals, cells have evolved eukaryotic initiation factor 5A (eIF5A), bearing a unique hypusination modification, to facilitate peptide-bond formation. Although initially thought to act primarily at poly-proline motifs (Gutierrez et al., 2013; Lassak et al., 2015; Huter et al., 2017), later studies have suggested a broader role for hypusinated eIF5A (h-eIF5A) in translation elongation (Schuller et al., 2017; Pelechano and Alepuz, 2017; Manjunath et al., 2019). eIF5A therefore represents a key factor in modulating context-dependent translation elongation, making it a prime target for exploring the interplay between coding sequence, elongation, and initiation.

A conceptual framework commonly used to model translation is the Totally Asymmetric Exclusion Process (TASEP), where ribosomes are treated as particles stochastically hopping along the mRNA (modeled as a 1D lattice) (Zia et al., 2011; Sharma et al., 2019; Szavits-Nossan and Ciandrini, 2019; Erdmann-Pham et al., 2020). In this framework, entry and exit rates correspond to initiation and termination, respectively, while the hopping rate represents elongation. Under conditions where initiation is rate-limiting, TASEP predicts a “low-density” regime featuring sparse ribosome loading and minimal collisions (Riba et al., 2019).

While the TASEP provides a strong theoretical basis for understanding translation, extending it to capture single-mRNA heterogeneity and dynamic fluctuations in initiation and elongation is a persistent challenge (Zia et al., 2011; Szavits-Nossan et al., 2018; Andreev et al., 2018; Levin and Tuller, 2018). Advances in single-molecule imaging, particularly the SunTag system, enable real-time monitoring of multiple rounds of translation on individual mRNAs (Tanenbaum et al., 2014; Yan et al., 2016). However, most analyses of SunTag data rely on averaging signals across many transcripts, thereby masking transcript-to-transcript variability (Morisaki and Stasevich, 2018; Aguilera et al., 2019). While a model that assigns distinct initiation and elongation rates to each transcript would offer deeper insights, this is complicated by the fact that the fluorescence of one translation site is the sum of the signals of multiple translating ribosomes. Because the fluorescence intensity is constant after the SunTag and the noise is usually large, inferring the position of single ribosomes on the mRNA is extremely challenging.

In this study, we develop a framework that decodes individual single-mRNA traces while simultaneously estimating global initiation and elongation parameters, allowing us to leverage single-molecule resolution without requiring explicit knowledge of ribosome positions on each transcript. We applied our model to investigate how CDS features modulate translation elongation and initiation in mammalian cells, using single-molecule (SunTag) imaging in HeLa cells (Weidenfeld et al., 2009; Voigt et al., 2017; Tanenbaum et al., 2014). Specifically, we designed multiple SunTag reporters differing only in codon content in the second half of the CDS, comparing proline-rich sequences such as those of collagen type I alpha 1 (COL1A1), highly dependent on h-eIF5A (Barba-Aliaga et al., 2021), to minimal or alanine-rich inserts. We acquired SunTag traces with and without harringtonine treatment. Our TASEP-based model allowed us to infer reporter-specific translation kinetics and ribosome number over time for each trace. We further exploit the role of eIF5A by perturbing elongation via pharmacological inhibition and genetic knockout, thereby examining the effects on ribosome density and initiation. Overall, our study contributes new tools for analyzing SunTag data and provides broader insights into how translational parameters are dynamically regulated to maintain protein synthesis homeostasis.

Results

Translation exhibits bursts with low average ribosome density

To image translation of single mRNAs over time in living cells, we used the SunTag system (Tanenbaum et al., 2014), which fluorescently tags nascent protein chains on individual transcripts. The SunTag encodes an array of 24 GCN4 epitopes recognized by single-chain variable fragment antibodies fused to a super-folder GFP (scFv-GFP). When this tag is placed upstream of a sequence of interest, newly synthesized peptides emerging from the ribosome exit tunnel are rapidly bound by scFv-GFP (Yan et al., 2016). The mRNA itself is fluorescently labeled with an MS2-MCP system (Figure 1A).

To investigate the influence of coding sequence on translation elongation and initiation, we derived multiple reporters from the SunTag-Renilla-MS2 system (Voigt et al., 2017), substituting the Renilla sequence with different inserts (Figure 1A, B). We designed a proline-rich reporter (PPG) containing a COL1A1 subsequence rich in proline-proline-glycine (P/P/G) repeats, which are known to induce ribosome stalling and depend on eIF5A activity (Barba-Aliaga et al., 2021; Gutierrez et al., 2013; Schuller et al., 2017; Pelechano and Alepuz, 2017). As controls, we generated two additional reporters: one in which the proline codons in the COL1A1 subsequence were replaced with alanine (AAG), a neutral amino acid encoded by fast-translating codons (Gobet et al., 2020), to minimize stalling and possible eIF5A dependence; and another with no insert sequence (no-insert) as a control for SunTag protein synthesis. The original SunTag-Renilla-MS2 reporter (Renilla) was also included in our analysis (Figure 1B). Together, these four constructs allowed us to systematically compare the translation dynamics of mRNAs with varying coding sequences and to evaluate the impact of elongation perturbation on translation kinetics.

All reporters were stably integrated into HeLa cells under a doxycycline-inducible promoter (Weidenfeld et al., 2009). The cell line also stably expresses scFv-GFP antibodies against the SunTag together with NLS-stdMCP-stdHalo-RH1 recognizing the MS2 loop (MCP) while binding to the actin cortex (RH1), stabilizing the mRNA for long-term imaging (Bhaskar et al., 2020) (Figure 1C).

Using a spinning-disk confocal microscope, we imaged translation over several minutes (Figure 1—figure Supplement 1A and Figure 1—video 1), observing a highly dynamic and heterogeneous process characterized by active and inactive periods (Figure 1D). We quantified the duration of these periods empirically, by measuring the duration of silent (inactive) and translated (active) periods. In most experiments, more than 50% of mRNAs were translated (Figure 1E) with a mean trace duration exceeding 10 minutes (Figure 1—figure Supplement 1B). We quantified the duration of translated and silent periods (Figure 1F), finding that active translation typically lasts around 10–15 minutes, interspersed with shorter silent periods of approximately 5–10 minutes across all reporters (Figure 1G). The slightly shorter translated periods observed for the no-insert reporter are likely due to its shorter length, which allows observation of isolated bursts more frequently.

We established an experimental method to measure the intensity of one mature protein, allowing us to estimate the number of translating ribosomes across translated periods. The measurement of mature proteins is challenging due to their rapid diffusion and low intensity against the high background of unbound scFv-sfGFPs. To overcome this, we used a SunTag-Renilla-RH1 construct to tether mature proteins to the actin cytoskeleton (Voigt et al., 2017). However, as these spots remained too dim at experimental laser powers, we developed an intensity-power calibration assay. This allowed us to measure spot intensity at high power and extrapolate the value for

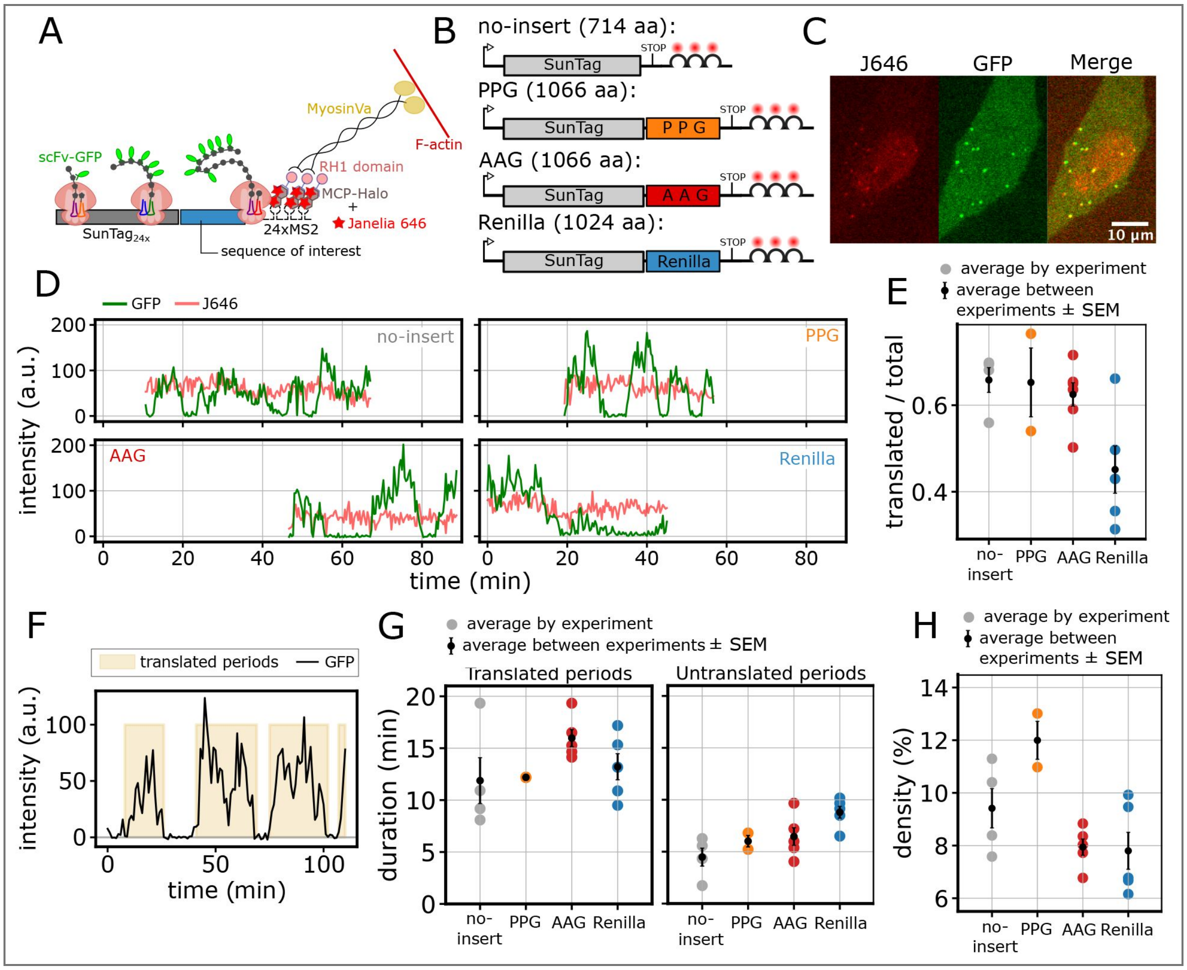

Figure 1. **Translation exhibits bursts with low average ribosome density.**

(A) Schematic of the SunTag system used for single-molecule imaging of translation. **(B)** SunTag reporters differing in their coding sequence inserts: no insert, PPG (proline-rich), AAG (alanine-rich) and Renilla. **(C)** Representative live-cell image of a HeLa cell expressing the PPG reporter, showing JF646 (mRNA), GFP (nascent peptide), and merged signals. The image is part of a larger field of view, acquired with a 60X objective in a spinning-disk confocal setup. Scale bar, 10 µm. **(D)** Representative long traces for each reporter, showing stable JF646 (mRNA, red) and fluctuating GFP (SunTag, green) intensities over time. **(E)** Average fraction of translated traces for each reporter. Colored dots represent averages from individual experiments; black dots with bars show the mean ± SEM across multiple experiments (*n* traces, *n* experiments): no-insert (669, 4), PPG (440, 2), AAG (1235, 6), Renilla (804, 5). **(F)** Example of the SunTag signal of a translated mRNA (no-insert), highlighting translated periods (shaded yellow). **(G)** Average duration of translated periods (left) and untranslated periods (right) for traces longer than 20 min. Colored dots represent averages from individual experiments; black dots with bars show the mean ± SEM across multiple experiments (*n* traces, *n* experiments): no-insert (41, 4), PPG (45, 2), AAG (90, 5), Renilla (62, 5). **(H)** Estimated average ribosome density (percentage of transcript covered by ribosomes) for traces in (G). Significance tests performed with Mann-Whitney U test. Figure 1—figure supplement 1. SunTag traces in control conditions. Figure 1—figure supplement 2. Experimental measurement of mature protein intensities to estimate number of translating ribosomes.

liveimaging conditions (14 ± 2 a.u., Methods and Figure 1—figure Supplement 2A–C). No-insert, AAG and Renilla reporters showed on average less than 10 ribosomes per mRNA (Figure 1—figure Supplement 2D), while the PPG reporter showed a slightly higher number (12 ribosomes per mRNA).

Assuming a uniform ribosome distribution along the transcript during a translated period, these observations imply a low ribosome density, with 7-12% of the transcript length covered by ribosomes, on average (Figure 1H). However, given the observed bursting dynamics, the actual ribosome distribution may deviate from a uniform distribution, potentially resulting in higher local ribosome densities. Nevertheless, our data show that, independently of the reporter sequence, bursting dynamics and ribosome densities were very similar, although this does not inform on the relative contributions of initiation and elongation to these observations. Finally, the observation of bursting behavior is in agreement with what was measured Livingston et al. (2023) for other SunTag reporters.

TASEP-based inference of translation dynamics from single run-off traces

To gain more information on the translation dynamics, we performed harringtonine (HT) run-off assays, under the same imaging conditions (Figure 2—figure Supplement 1A, Figure 2—video 1 and Figure 2—video 2). HT blocks the first step of elongation after subunit joining, while allowing translating ribosomes to elongate and terminate (Fresno et al., 1977). The time required for the SunTag signal to disappear approximates the total elongation time, but depends on the position of the last (most upstream) elongating ribosome when HT blocks initiation. Because the distribution of ribosomes on each transcript is unknown, it is common practice to average the signal from multiple traces to estimate the total elongation time (hence, the average ribosome speed) (Yan et al., 2016; Wang et al., 2016; Mateju et al., 2020; Aguilera et al., 2019). Although this pseudobulk approach has been successful to estimate average elongation times, it has the limitation of neglecting trace-to-trace heterogeneity.

Here we used a novel approach, allowing us to infer the number of translating ribosomes over time at the single-mRNA level. In particular, we used a Hidden Markov Model (HMM), where the hidden states represent the number of ribosomes translating the SunTag reporter during the runoff (Figure 2A and Methods). The transition probabilities between hidden states were derived from an approximation of the Totally Asymmetric Exclusion Process (TASEP), assuming low ribosome density and homogeneous elongation rates. These probabilities define the likelihood of observing one or more termination events per imaging time step. This approach allowed us to relate the observed run-off times to the underlying initiation (α) and elongation rates (λ) (Figure 2A), as well as to decode single traces by inferring the number of translating ribosomes over time (Figure 2B). We assumed that HT takes 60 seconds to diffuse into the cells and effectively block initiation (Ingolia et al., 2009; Aguilera et al., 2019).

In our model, we accounted for the finite size of the SunTag by calculating correction factors for the intensity mean and variance ($\gamma(t)$ and $v(t)$, Methods). In particular, $\gamma(t)$ generalizes the timeindependent correction factor previously derived by others (Aguilera et al., 2019) to the run-off dynamics (Figure 2C and Methods). We first tested these approximations by comparing them with numerical simulations of the homogeneous ℓ-TASEP (Methods). We used different initiation and elongation rates, corresponding to a broad range of ribosome densities. These simulations confirmed the validity of our analytical approximations at low ribosome densities (Figure 2C, D and Figure 2—figure Supplement 2A).

We then tested the performance of the model on simulated SunTag traces, with different values of the kinetic parameters and ribosome density (ρ = 0.01 − 0.5, i.e. average fraction of occupied transcript) (Figure 2E). Instead of fixing the mature protein intensity i_{MP}, we used the model to infer its value, alongside α and λ. This parameter determines the extent of the intensity drop due to a termination event. The kinetic parameters and the mature protein intensity are accurately inferred at low density ($\rho \lesssim 10\%$) for different noise levels (Figure 2E), despite some correlation

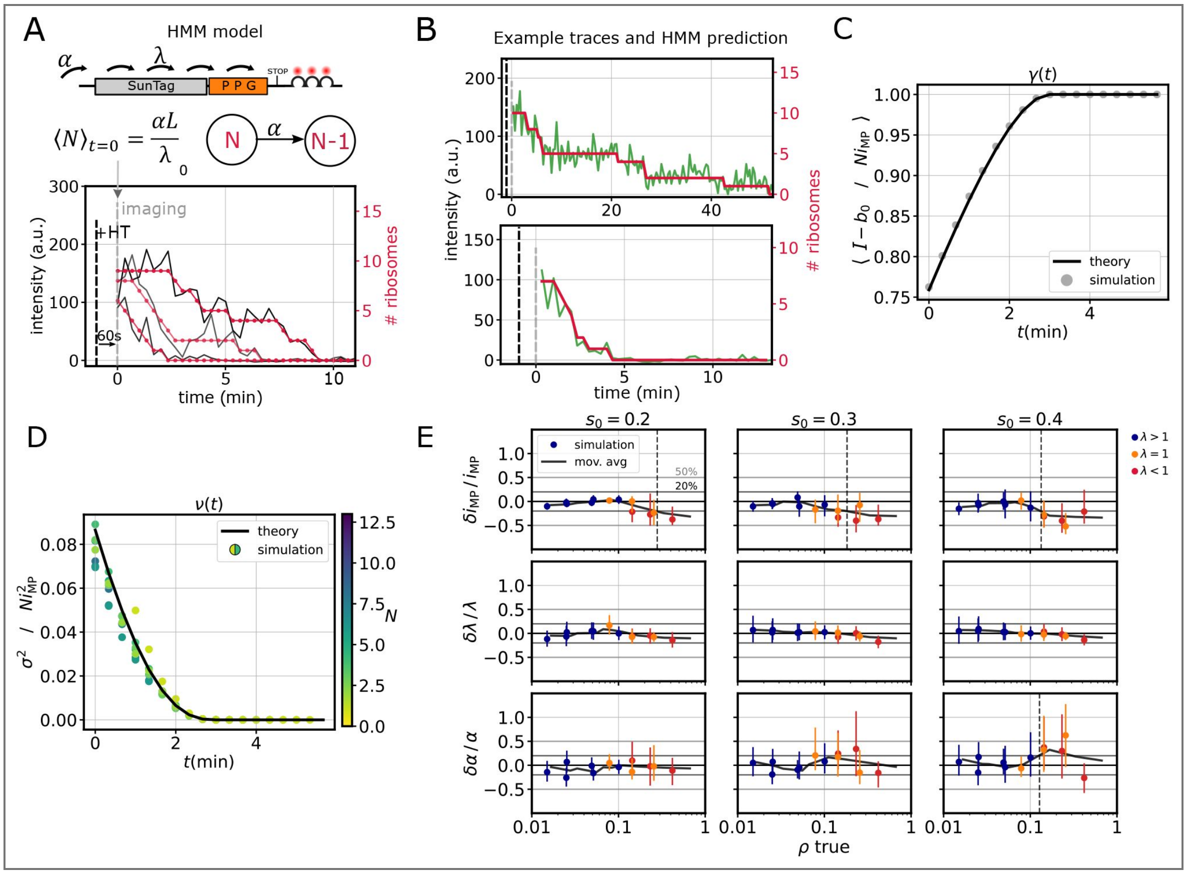

Figure 2. **TASEP-based inference of translation dynamics from single run-off traces.**

(A) Schematic of the Hidden Markov Model (HMM) used to analyze run-off experiments. α = initiation rate, λ = average elongation rate, L = reporter length, $\langle N \rangle_{t=0}$ = average number of ribosomes at $t = 0$ (60 s after HT addition). **(B)** Representative slow (top) and fast (bottom) decaying traces from one run-off experiment with the PPG reporter; intensity in green and predicted number of ribosomes in red. **C)** Comparison between the analytical correction factor γ (black line, Methods Equation 24) and the results obtained by numerical simulations of run-off traces (grey dots). The latter represent $(I(N,t) - b_0)/Ni_{\mathrm{MP}}$ averaged over 2000 simulated run-off traces, with $I(N,t)$ simulated intensity, b_0 offset, N number of ribosomes and i_{MP} mature protein intensity. Kinetic parameters used in simulations: initiation $\alpha = 1/60\ \mathrm{s}^{-1}$, elongation $\lambda = 3.0$ aa/s. **D)** Comparison between the analytical correction factor ν (black line, Methods Equation 26) and the results obtained by numerical simulations of run-off traces (coloured dots). The latter represent the normalized intensity variance $\sigma^2(N,t)/Ni_{\mathrm{MP}}^2$ for different values of N, same simulated traces as in C). **E)** Relative error on model parameters (α, λ, i_{MP}) *vs* average ribosome density ρ, for different values of the noise parameter s_0 (Methods Equation 44), for simulated run-off traces. Each data point is obtained with a different combination of $\lambda \in \{0.5, 1, 3, 5\}$ aa/s and $\alpha \in \{1/120, 1/60, 1/30\}\ \mathrm{s}^{-1}$. Blue dots represent the relative error on each parameter, the black curve is a moving average, and the dashed line indicates the density ρ at which the relative error goes beyond 0.2. Simulation parameters in C) – E): i_{MP} = 10 a.u., reporter length L = 1066 aa, ribosome size ℓ = 10 aa. Figure 2—figure supplement 1. SunTag run-off traces in control conditions after HT addition. Figure 2—figure supplement 2. TASEP-based inference accurately estimates ribosome density on simulated data despite correlations between α and i_{MP}.

between α and i_{MP} (Figure 2—figure Supplement 2B). The correlation is more pronounced at higher density, where i_{MP} is underestimated and α is overestimated. Importantly, the average ribosome density is overestimated at high densities – we are neglecting ribosome interference that can reduce the effective initiation rate – while it is accurately inferred at low densities (Figure 2—figure Supplement 2C).

Low ribosome density arises from coordinated translation initiation and elongation

After testing our model on simulated traces, we used it to analyze the HT run-off assays. To avoid correlations between parameters, we fixed the value of i_{MP} to the experimentally measured value (14 ± 2 a.u.). We modeled intensity noise as lognormal and estimated its magnitude from cycloheximide-treated traces (Methods and Figure 3—figure Supplement 1). The model allowed us to estimate the number of ribosomes translating the mRNA for each translation site, and thus the total run-off time for each trace (i.e., the time at which the number of translating ribosomes reaches zero) (Figure 3A, violin plot). These single-trace estimates revealed substantial variability in run-off times, which likely reflects elongation heterogeneity and stalling events beyond what can be explained by differences in initial ribosome distribution.

However, for some traces, the mRNA signal was lost before run-off completion (Figure 3A, histogram). This occurred rarely for the no-insert reporter, while for the Renilla and PPG reporters, we observed about 10-20% of incomplete traces. Surprisingly, the AAG reporter showed up to 50% of incomplete traces per experiment and longer run-off times, suggesting considerable ribosome stalling (Figure 3A). The elongation rates inferred by our model for the no-insert, PPG and Renilla reporters ranged from 2 to 4.5 aa/s, consistent with previous reports (2–6 aa/s) (Yan et al., 2016; Mateju et al., 2020; Livingston et al., 2023; Madern et al., 2025) (Figure 3B). As we expected from the run-off times distribution, the AAG reporter was markedly slower, with elongation rates close to 1 aa/s (Figure 3B).

It is common practice (Yan et al., 2016; Wang et al., 2016; Mateju et al., 2020; Aguilera et al., 2019) to infer the total elongation time by fitting a linear decay to the average run-off intensity, usually discarding the slower decay resulting from heterogeneity in elongation (Wang et al., 2016). The intercept between the fast linear decay and the x-axis gives an estimate of the total average elongation time (Figure 3—figure Supplement 2A). Our single-trace inference method tends to estimate lower elongation speeds with respect to what was obtained with the population-averaged linear fit (Figure 3—figure Supplement 2A–B). This is likely because our model accounts for the initial ribosome distribution and is more sensitive to trace-to-trace translation heterogeneity. In addition, the HMM allowed us to decode the single-mRNA traces and estimate the time between consecutive termination events, which significantly deviated from an exponential distribution (Figure 3—figure Supplement 2C). This can be partially explained by bursting; however, since initiation is inhibited upon HT treatment, waiting times between termination events that are much longer than the average elongation time most likely reflect persistent ribosome stalling.

The inferred initiation rates corresponded to ∼1-2 ribosomes per minute (Figure 3C), close to what was measured in similar systems (1 – 5 ribosomes per min) (Boersma et al., 2019; Barrington et al., 2023). Surprisingly, the AAG reporter showed markedly lower initiation rates (Figure 3C). From the elongation and initiation rates, we calculated the average ribosome density under the low-density assumption (Figure 3D, Methods Equation 14). Despite the variability in elongation rates, we obtained low densities for all reporters, remarkably similar to our previous estimates (Figure 1H). Furthermore, we observed a significant correlation between initiation and elongation rates (Pearson correlation coefficient $r = 0.85$, average density $\rho = 0.08$) (Figure 3E). This suggests a tight feedback between initiation and elongation, where changes in one rate are compensated by changes in the other to maintain a low ribosome density.

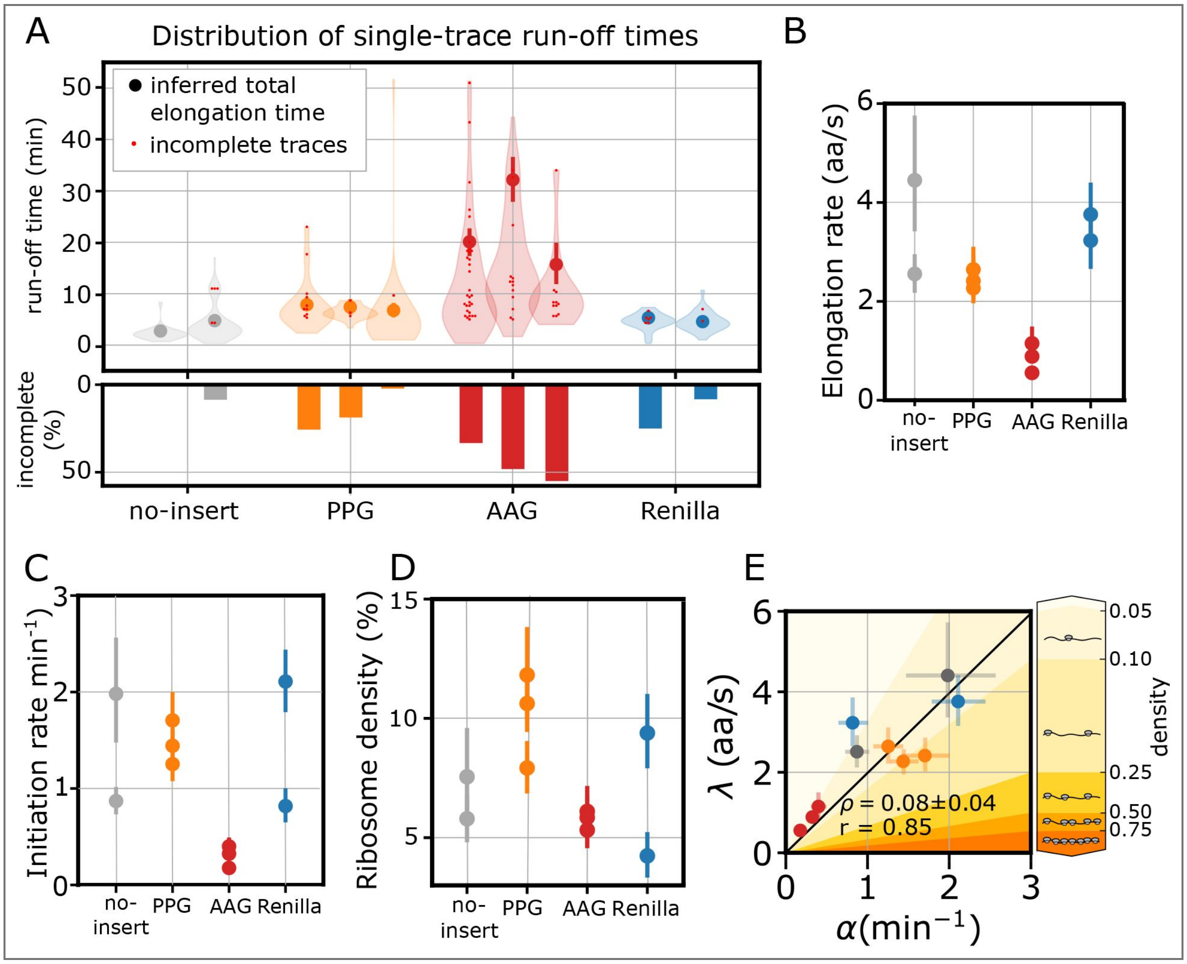

Figure 3. Low ribosome density arises from coordinated translation initiation and elongation.

In all panels the error bars indicate 95% confidence intervals. **(A)** Run-off time distribution (top) and percentage of incomplete run-offs (bottom) for each reporter. For incomplete runoffs we include the total duration of the trace, as it represents a lower-bound to the total run-off time (red dots). **(B)** Inferred elongation rates (λ). **(C)** Inferred initiation rates (α). **(D)** Inferred average ribosome density. **(E)** Correlation between elongation and initiation rates for each experiment. The shaded background indicates approximate ribosome density regimes, ranging from low density (light yellow) to high density (dark yellow), as shown by the adjacent color bar. ρ indicates the average density ± standard deviation, r is the Pearson correlation coefficient. Figure 3—figure supplement 1. Global measurement noise estimation from CHX traces. Figure 3—figure supplement 2. HMM approach predicts lower elongation speed than simple linear regression and suggests the presence of bursting and stalling.

eIF5A perturbations minimally affect ribosome density and translational bursting

Having established a correlation between initiation and elongation rates across reporters, we next aimed to investigate the impact of perturbing translation elongation on initiation and ribosome density. We chose to target eIF5A, a protein known to play a crucial role in ribosome elongation (Schuller et al., 2017; Pelechano and Alepuz, 2017; Manjunath et al., 2019). eIF5A activity was modulated using two distinct approaches. First, HeLa cells were incubated for 24 hours with 1 μM and 10 μM GC7. GC7 is a spermidine analog that inhibits Deoxyhypusine synthase (DHPS), the enzyme responsible for hypusination of eIF5A, thereby reducing the levels of active, hypusinated eIF5A (h-eIF5A) (Barba-Aliaga et al., 2021; Giraud et al., 2020; Schultz et al., 2018; Coni et al., 2020; Szepesi et al., 2023; Matsumoto et al., 2023). Upon treatment, h-eIF5A levels were reduced by ∼ 2and ∼ 3-fold, respectively, while total eIF5A levels remained unchanged (Figure 4A). To minimize potential side effects of GC7, we selected the 1 μM concentration throughout our experiments. As a second approach, we generated a CRISPR/Cas9 eIF5A knockout (KO) cell line derived from the PPG-expressing HeLa cells. This KO cell line exhibited a complete loss of both total and hypusinated eIF5A (Figure 4B).

Upon doxycycline (DOX) induction, both GC7-treated cells and eIF5A KO cells showed expression of the SunTag reporters and exhibited active translation (Figure 4C–E, Figure 4—figure Supplement 1A, Figure 4—video 1 and Figure 4—video 2), with no significant difference in the fraction of translated mRNA (Figure 4F) and translated trace duration (Figure 4—figure Supplement 1B). As for control conditions, the translation signal showed bursting dynamics (Figure 4E), with duration of translated and untranslated periods similar to the control (Figure 4G, H). Ribosome density was remarkably stable between control and perturbed conditions, with no statistically significant differences (Figure 4I). In summary, these findings indicate that decreasing eIF5A activity or level has a minimal effect on translational bursting and ribosome density in our system.

eIF5A-driven elongation changes induce initiation regulation to preserve ribosome density

To understand how elongation and initiation contribute to maintaining low ribosome density upon eIF5A perturbations, we performed HT run-off experiments on GC7-treated and eIF5A KO cells, alongside control conditions (Figure 5A, Figure 5—figure Supplement 1A, Figure 5—video 1, Figure 5—video 2 and Figure 5—video 3).

Using our TASEP-based model, we evaluated the intensity of a single mature protein by jointly fitting control and perturbed conditions with a shared i_{MP} parameter, while still allowing conditionspecific initiation and elongation rates (Figure 5—figure Supplement 2A). Strikingly, the inferred i_{MP} values closely matched the independently measured intensity of (14 ± 2) a.u. (Figure 5—figure Supplement 2B). This close quantitative agreement, combined with the absence of any significant correlation between α and i_{MP} (Figure 5—figure Supplement 2C), indicates that the model recovers this parameter robustly rather than compensating for other rates. Given this strong internal consistency between measurement and inference, we fixed i_{MP} to its experimentally determined value for all subsequent analyses.

After 24 hours of GC7 treatment, we observed an increase in the average run-off time across all reporters (Figure 5A). The subset of traces showing slow run-off (≥15 min) – < 4% in control conditions, with the exception of the AAG – increased significantly upon GC7 treatment (11% for no-insert, 18% for PPG and 40% for Renilla) (Figure 5B). The average total elongation time inferred with our model also increased for all reporters (Figure 5C). The extent of the increase varied across experiments for no-insert and Renilla: 2 and 8-fold changes for no-insert and 8 and 5-fold changes for Renilla. More consistent although milder increases were observed for PPG, both upon GC7 treatment (2-fold change) and eIF5A KO (1.5-fold). The already slow AAG reporter was only mildly affected (1.4-fold change). Our HMM inference results confirmed a general decrease in

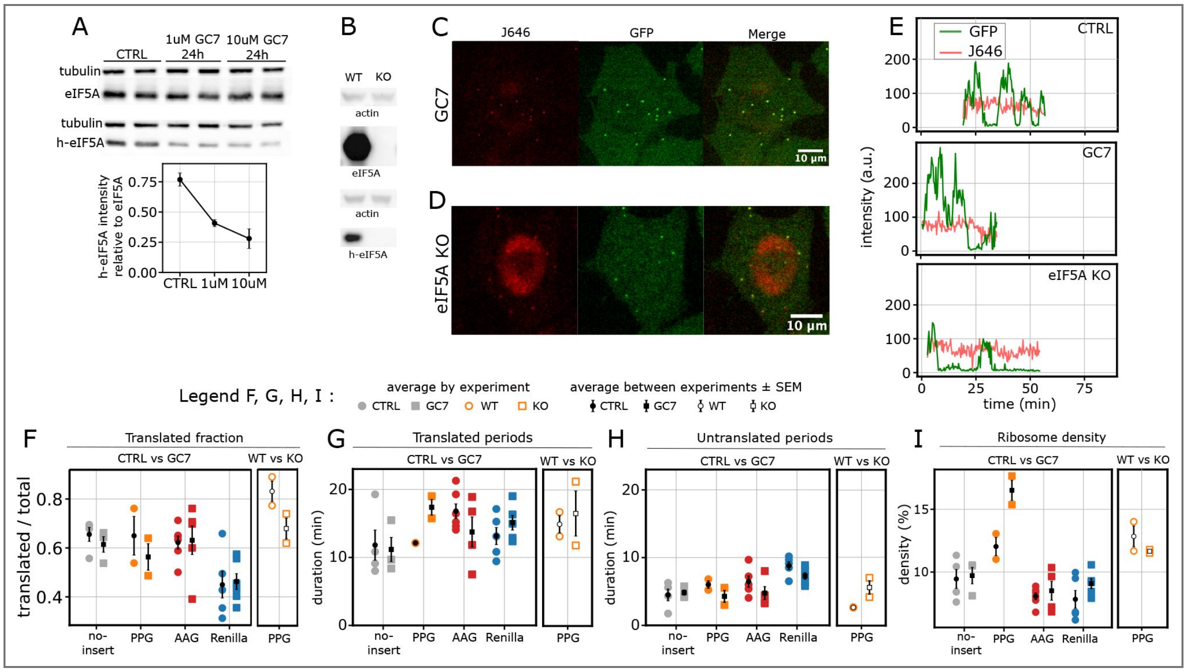

Figure 4. **eIF5A perturbations minimally affects ribosome density and translational bursting.**

(A) Top: western blot of h-eIF5A levels in control cells and cells treated with 1 µM and 10 µM GC7 for 24 hours, with 1 mM AG. Bottom: quantification of h-eIF5A signal relative to total eIF5A. Error bars represent standard deviation between replicates. **(B)** Total eIF5A and h-eIF5A expression in CRISPR/Cas9 knock-out (KO) clones compared to wild-type (WT) HeLa cells. **(C)** Representative live-cell images of a HeLa cell expressing the PPG reporter after 24-hour treatment with 1 µM GC7 and 1 mM AG, showing JF646 (mRNA, red), GFP (SunTag, green), and merged signals. The image is part of a larger field of view, acquired with a 60X objective in a spinning-disk confocal set-up. Scale bar, 10 µm. **(D)** Same as (C) for eIF5A KO cells. **(E)** Representative traces of the PPG reporter under control conditions, GC7 treatment, and eIF5A KO, showing GFP (SunTag, green) and J646 (mRNA, red) intensities over time. **(F)** Fraction of translated mRNAs (*n* traces, *n* experiments): no-insert CTRL (669, 4), GC7 (511, 3); PPG CTRL (440, 2), GC7 (392, 2); AAG CTRL (1235, 6), GC7 (900, 5); Renilla CTRL (804, 5), GC7 (1476, 6); PPG WT (396, 2), KO (343, 2). **(G)** Average duration of translated periods and **(H)** untranslated periods for traces longer than 20 min. **(I)** Average ribosome density during the translated periods analyzed in (G). **(G – I)** (*n* traces > 20 min, *n* experiments): no-insert CTRL (41, 4), GC7 (20, 3); PPG CTRL (45, 2), GC7 (28, 2); AAG CTRL (90, 5), GC7 (26, 4); Renilla CTRL (62, 5), GC7 (112, 5); PPG WT (33, 2), KO (26, 2). Colored dots represent averages from individual experiments, black dots with bars show the mean ± SEM across multiple experiments. Mann-Whitney U significance test. Figure 4—figure supplement 1. SunTag traces in perturbed conditions.

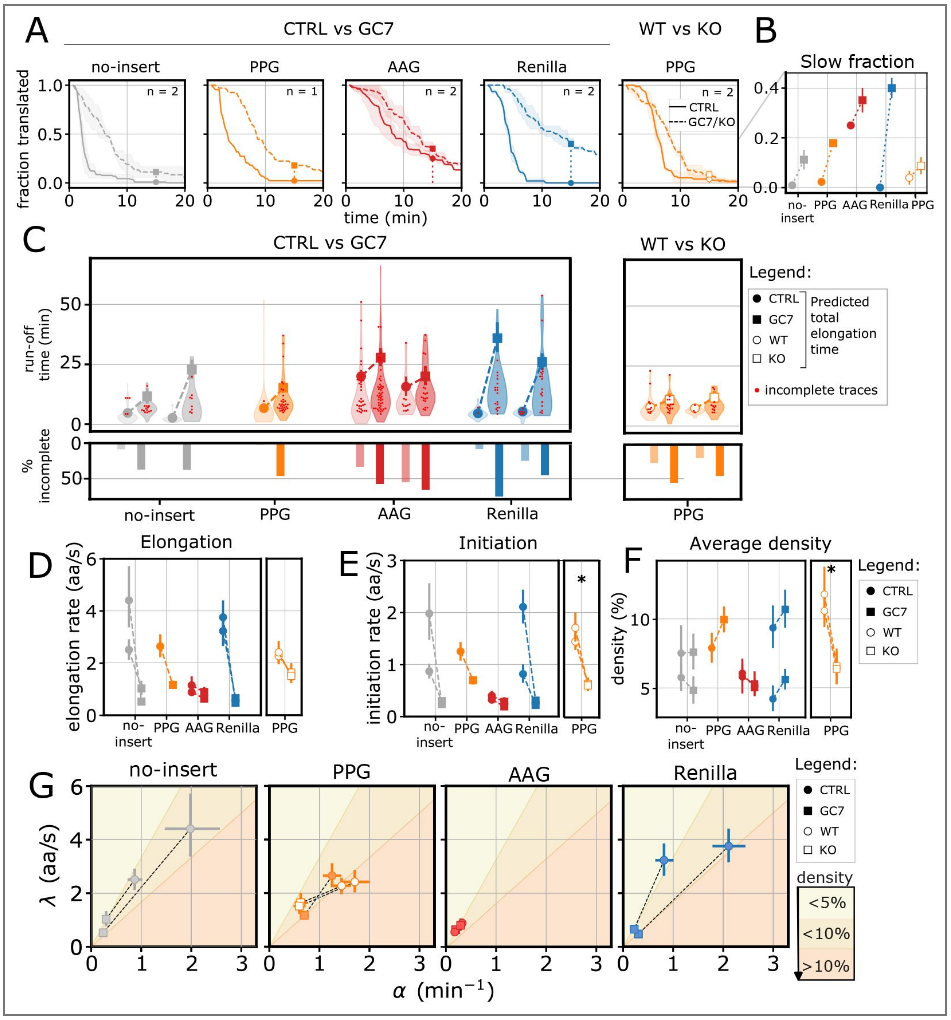

Figure 5. eIF5A-driven elongation changes induce initiation regulation to preserve ribosome density.

(A) Fraction of translated mRNAs over time in control (solid lines) and perturbed conditions (GC7 treatment and eIF5A KO, dashed lines) for each reporter. Thick lines represent the mean across experiments, shaded areas represent standard deviation between experiments. Full circles and squares represent the fraction of mRNAs still translated 15 min after HT addition (control and perturbed conditions, respectively). (*n* is the number of experiments) **(B)** Fraction of translated traces showing slow run-off (still translated after 15 min of HT treatment). **(C)** Run-off time distribution (top) and percentage of incomplete run-offs (bottom) for each reporter in control conditions (circles) and perturbed conditions (GC7 treatment and eIF5A KO, squares). For incomplete runoffs we include the total duration of the trace, as it represents a lower-bound to the total run-off time (red dots). Number of translated traces (control, perturbed): no-insert (58, 35), (22, 24); PPG (45, 67); AAG (93, 80), (20, 32); Renilla (24, 24), (24, 40); PPG WT/KO (39, 28), (16, 25). **(D)** Inferred elongation rates. **(E)** Inferred initiation rates. **(F)** Inferred average ribosome density. **(G)** Representation of changes in translation kinetic parameters upon eIF5A perturbation in the initiation-elongation ($\alpha - \lambda$) plane. The shaded background indicates approximate ribosome density regimes, ranging from low density (light yellow) to high density (dark yellow), as shown by the adjacent color bar. Significance tests performed with Mann-Whitney U test. Figure 5—figure supplement 1. SunTag run-off traces in perturbed conditions after HT addition. Figure 5—figure supplement 2. Inferred mature protein intensities are in good agreement with the experimental measurement.

elongation rates in both GC7-treated and eIF5A KO cells (Figure 5D). Importantly, we also observed a concurrent decrease in initiation rates under both perturbation conditions (Figure 5E).

In GC7-treated cells, both elongation and initiation rates decreased proportionally (Figure 5D,E), resulting in a nearly constant ribosome density (Figure 5F, G). This is consistent with our previous observations (Figure 4I), which showed that average ribosome density was largely unaffected by the GC7 treatment. However, the eIF5A KO cells revealed a different outcome. While elongation rates also decreased in the KO, the magnitude of the decrease was less pronounced compared to GC7 treatment (Figure 5D). In particular, the fraction of traces showing a slow run-off (Figure 5A, right tail, and Figure 5B) was reduced in eIF5A KO with respect to GC7 treatment (18% in GC7 treatment and 9% in eIF5A KO). At the same time, initiation was strongly decreased (Figure 5E). As a result, we observed a significant reduction in average ribosome density in the eIF5A KO cells (Figure 5F, G, 30–40% decrease), which we did not observe upon GC7 treatment. This can be explained by either faster elongation or more efficient stalling resolution via RQC mechanisms (Latallo et al., 2023; Goldman et al., 2021). Overall, we observed that initiation is dynamically adjusted to elongation upon eIF5A perturbation, most likely to avoid ribosome crowding.

Discussion

By integrating single-mRNA imaging with single-trace statistical modeling, our study highlights key principles of translational regulation. First, we demonstrate that translation maintains remarkably low ribosome density (≤ 12% of the CDS occupied by ribosomes) across diverse coding sequences by coordinating initiation and elongation rates. Second, when elongation is perturbed by modulating eIF5A activity, initiation is proportionally adjusted to preserve low-density translation. Third, complete eIF5A knockout (KO) alters this coordination, suggesting differences in the sensing of ribosome stalling or collisions in the absence of eIF5A.

Our results support a model in which translation under homeostatic conditions occurs at low ribosome density, with initiation serving as the rate-limiting step (Riba et al., 2019; Boersma et al., 2019; Livingston et al., 2023). Indeed, using the SunTag system to visualize translation of single mRNAs, we consistently observed low density across all reporter mRNAs, regardless of coding sequence variations. This conserved feature may reflect an evolutionary strategy aimed at avoiding overuse of resources, as the same protein synthesis rate can be attained by engaging fewer ribosomes at the same time (Erdmann-Pham et al., 2020). Since the pool of free ribosomes may limit initiation (Shah et al., 2013), maximizing this pool may be beneficial for the cell, since it allows more rapid and flexible adaptation of initiation rates in response to changing demands. At the same time, by minimizing the frequency of transient, stochastic collisions that arise from high ribosome density, the cell enhances its ability to detect and respond to problematic translation events. Persistent ribosome collisions can serve as clear signals of aberrant translation and trigger mRNA degradation, in agreement with the observation that high ribosome occupancy can destabilize mRNAs (Presnyak et al., 2015; Bazzini et al., 2016; Narula et al., 2019; Wu et al., 2019; Bae and Coller, 2022; Bicknell et al., 2024).

The low ribosome density across reporters stems from the dynamic interplay between elongation and initiation rates. Even though our mRNAs shared an identical 5’ UTR, we observed significant and coupled variations in elongation and initiation across different coding sequences. It is important to note that this correlation is not a consequence of our model or inference, as the results on simulated data showed that we could infer a wide range of initiation and elongation rates, corresponding to very different densities. The strong correlation observed in real data suggests a feedback mechanism in which initiation is modulated by the prevailing elongation status, thereby preventing ribosome crowding and maintaining low density. The dynamic interplay between elongation and initiation has also recently been suggested in yeast (Lyons et al., 2023), in the case of mRNA reporters encoding synonymous codons. Similar conclusions have been reached in flies and humans for synonymous reporters, pointing to feedback mechanisms on translation initiation (Barrington et al., 2023). The main hypothesis underlying these observations in mammals is that suboptimal elongation leads to ribosome collisions, which are in turn sensed via molecular pathways that inhibit initiation.

The first response to ribosome collisions is the recruitment of the GIGYF2/4EHP complex, a cisacting translational repressor that inhibits eIF4E binding and thus blocks further initiation (Hickey et al., 2020; Juszkiewicz et al., 2020). The GIGYF2/4EHP complex can rapidly repress initiation upon the detection of collisions (whereas more persistent collisions initiate RQC, e.g. ZNF598-mediated degradation) (Wu et al., 2020). This molecular feedback on initiation, being transcript-specific, could explain our observations, contrary to a more general response to collisions such as the Integrated Stress Response (ISR) and eIF2α phosphorylation, which would globally attenuate initiation (Wu et al., 2020). It is important to note that our reporters differ in codon content only far downstream of the translation start site (about 600 aa), which allows us to exclude the hypothesis of ribosomes queuing up to the initiation site and physically blocking initiation. Indeed, this would result in a high ribosome density, in contrast to our observations.

If collisions are long-lived, alternative molecular pathways, for example involving ZNF598, can induce RQC and, eventually, mRNA degradation. Also, beyond ribosome collisions, other mechanisms were shown to detect slow-elongating ribosomes in yeast, inducing ribosome recycling and mRNA degradation (Buschauer et al., 2020; Li et al., 2022) – the evidence of similar pathways in mammals is still limited (Absmeier et al., 2022). Although our reporters showed a broad range of elongation rates (1-4 aa/s), there was no evidence of translation-dependent mRNA degradation, as we observed relatively consistent fractions of translated transcripts and trace duration distributions across reporters.

eIF5A plays a critical role in promoting elongation, especially in proline-rich sequences such as *COL1A1*. Thus, we next examined how perturbing eIF5A influences elongation, initiation, and ribosome density. Remarkably, significant reductions of active eIF5A – via GC7 treatment or genetic knockout – maintained largely consistent bursting dynamics and average ribosome densities compared to controls. However, run-off assays revealed different underlying mechanisms. GC7 treatment led to a coordinated decrease in both elongation and initiation rates, maintaining low ribosome density. Surprisingly, we observed stronger changes in run-off profiles for no-insert and Renilla compared to PPG, contrary to what was expected, as PPG contains known eIF5A-target motifs. It is possible that stronger stalling or collisions in PPG were masked by the RQC-mediated degradation of the SunTag reporter. Such degradation would lead to faster decay kinetics and could result in incorrect interpretation of the apparent translation dynamics (Latallo et al., 2023). Nevertheless, our perturbation experiment suggests that moderately impaired elongation due to reduced eIF5A activity triggers a feedback loop that dynamically adjusts initiation downwards, maintaining low-density translation.

eIF5A KO appeared to affect translation differently than GC7 treatment, leading to a significant (30–40%) reduction in ribosome density due to a stronger decrease in initiation than elongation. In particular, eIF5A KO showed fewer traces showing slow run-off compared to GC7, resulting in a milder decrease in elongation rate relative to control conditions. Although we do not have a definitive explanation for this observation, we hypothesize that the strong reduction in total eIF5A levels could alter the kinetics of stalling recognition and ribosome degradation. In yeast, eIF5A is known to compete with Not5 (CNOT3 in human (Absmeier et al., 2022)) for binding to the ribosomal E site (Buschauer et al., 2020). When peptidyl transfer is slow, eIF5A can occupy the E site and resolve stalling by promoting peptide-bond formation. Conversely, when decoding is slow, both the A and E sites remain unoccupied, allowing Not5 to bind the E site and mark the ribosome for degradation. Recently a similar mechanism was suggested in mammals (Absmeier et al., 2022). In this scenario, the absence of eIF5A could lead to more frequent or faster ribosome degradation, reducing the accumulation of stalled ribosomes and giving the appearance of a higher elongation rate. Ribosome recycling in normal conditions was recently estimated to take 22 min on average (Madern et al., 2025), but could be faster in the absence of eIF5A.

Interestingly, our attempt to optimize translation through proline-to-alanine substitutions in the COL1A1 PPG motif (AAG reporter) resulted in markedly slower elongation rates (~ 1 aa/s)) and increased frequency of incomplete run-off traces (50%). This suggests that expressing nonnative sequences, even with seemingly favorable substitutions and a comparable codon adaptation index, can introduce unanticipated challenges, such as impaired RNA secondary structure or co-translational folding constraints. Specifically, replacing prolines with alanines reduced elongation by 60%, likely due to disrupted co-translational folding. Given that prolines in collagen are critical for triple-helix formation (Shoulders and Raines, 2009), proline-to-alanine substitutions likely generate misfolded intermediates that stall ribosomes (Barba-Aliaga et al., 2021; Komar et al., 2024).

Limitations

First, our observations derive from exogenous mRNA, which may not fully recapitulate endogenous collagen translation dynamics, particularly given the absence of ER-coupled processing and secretory pathway targeting. This difference may explain the limited GC7 effects on the SunTag reporter, as recent studies suggest a direct role for eIF5A in ER-coupled collagen translation and translocation (Rossi et al., 2013; Mandal et al., 2016; Barba-Aliaga et al., 2021). This limitation could be partially overcome by genetically encoding the SunTag sequence via CRISPR/Cas9 genome editing, as performed by Dufourt et al. (2021) in Drosophila embryos. In addition, using an ER-specific SunTag reporter, as done in Choi et al. (2025) by adding a trans-membrane domain, would better recapitulate ER-coupled translation.

Second, while decoding individual traces, our model infers shared (population-level) rates. Inferring transcript-specific parameters would be more informative, but it is highly challenging due to the uncertainty regarding the initial ribosome distribution on single transcripts. Pooling multiple transcripts together allows some assumptions on the initial distribution that are essential for the inference of average elongation and initiation-rate parameters, while revealing substantial mRNAto-mRNA variability in the posterior decoding (e.g., Figure 3—figure Supplement 2C).

The deviation from an exponential distribution in the waiting times between termination events could arise from ribosome stalling, bursting, or a combination of both. Each of these effects would tend to bias our inference toward underestimating the true initiation rate. Incorporating explicit models of stalling and bursty initiation in future work would help make the inference framework more robust. In addition, our current model infers a single effective elongation rate per reporter, which averages ribosome dynamics across both the SunTag region and the variable CDS. As a consequence, it may underestimate true elongation-rate differences between the sequences of interest. This limitation could be alleviated by introducing distinct elongation parameters for the SunTag and the CDS, and by sharing the SunTag elongation parameter across different reporters. Incorporating all these features (stalling, bursting, SunTag-specific elongation rate) into the TASEP simulations would also make the simulated data more realistic.

Last, run-off assays are inherently variable and can underestimate or overestimate elongation rates, especially with uncertainties in harringtonine pharmacokinetics (Aguilera et al., 2019) or RQC-mediated termination events (Latallo et al., 2023; Goldman et al., 2021).

Outlook

An important contribution of our work is providing an initial framework for modeling translation kinetics at the single-trace level. Currently, this approach is tractable for run-off assays, where mRNAs eventually deplete their ribosomes, simplifying parameter estimation. Although challenging, extending the model beyond harringtonine assays would circumvent these limitations, while giving insights on initiation regulation, including bursting. This could be done by extending the model to steady-state translation and incorporating a 2-state telegraph model to capture “bursty” initiation. Furthermore, introducing constructs with different 5’UTRs, similar to the work of (Livingston et al., 2023), would provide valuable insights into how bursting dynamics interact with elongation rates. In addition, the current model infers population level parameters while accounting for single-trace stochasticity, extending the model (hierarchically) to allow for transcript-specific rates or stalling could provide valuable insights.

Addressing the current limitations, including the influence of RQC on run-off kinetics and the non-exponential distribution of termination events, will further refine our understanding of translation dynamics and improve model accuracy. In parallel, integrating run-off and bursting traces within a unified framework and incorporating transcript-specific kinetic parameters should offer a more complete picture of how cells regulate translation at the single-mRNA level. The latter could potentially be achieved by replacing the HMM with a particle-filter approach, to infer singletranscript kinetics (Wills and Schön, 2023). Finally, exploring the molecular mechanisms underlying the observed coupling of initiation and elongation, as well as the functional significance of bursting dynamics and ribosome density regulation, will be critical for unraveling how cells maintain proteostasis under diverse physiological and stress conditions.

Methods and Materials

Experimental Design

Model system and cell culture

We used a cell line derived from the HeLa-11ht cell line generated by Weidenfield et al. (Weidenfeld et al., 2009) which contains a site for Flp-RMCE (recombinase-mediated cassette exchange), allowing a single-copy genomic integration of a target gene, and constitutively expressing the reverse tetracycline-controlled transactivator (rtTA2-M2) for inducible expression. The cell line was further engineered to stably express GFP-tagged single-chain antibodies (scFv-GFP) against GCN4, and NLS-stdMCP-stdHalo-RH1 fusion protein (Voigt et al., 2017) (courtesy of Jeffrey Chao Lab, FMI Basel). The MCP protein recognizes a MS2 stem-loop cassette in the 3’ UTR of the reporter mRNA, while the RH1 domain associates with actin filaments through the interaction with the MyosinVa molecular motor allowing stable imaging of actin-anchored mRNAs (Figure 1A). In the following we refer to this cell line as the parental cell line, from which we obtain monoclonal cell lines expressing the SunTag reporters by stable transfection and fluorescence-activated cell sorting (FACS).

HeLa cells are cultured in Dulbecco’s Modified Eagle Medium (DMEM) containing 4.5 g/L glucose, Penicillin (100 U/ml), Streptomycin (100 μg/ml), L-Glutamine (4 mM) and 10% fetal bovine serum (FBS). Cells are maintained at 37°C and 5% CO2. For transient and stable transfections, we use Fugene HD Transfection Reagent (Promega) with Opti-MEM reduced serum medium (Thermo Fisher Scientific) according to the manufacturer’s instructions.

To perturb eIF5A activity, cells are treated with 1 μM N1-guanyl-1,7-diaminoheptane (GC7) for 24h before imaging. Because GC7 is known to be inactivated by the action of amine oxidases, which are abundant in serum, we add 1 mM aminoguanidine (AG) to inhibit amine oxidases, as is typically done in the literature (Maier et al., 2010). The control sample is treated with AG only.

DNA constructs

All the constructs described in this project are obtained from the previously described SunTagRenilla-MS2 plasmid (Wilbertz et al., 2019) (Addgene plasmid #119945, Figure 1A) via restrictionfree cloning. The resulting plasmids contain SunTag (24 x GCN4), destabilized FKBP domain (Banaszynski et al., 2006) with a stop codon and a 24xMS2 stem-loop cassette. In addition, PPG contains a subsequence of Human collagen type I α1 (transcript COL1A1-201, from nucleotide 2619 to 3672, 1056 nt), while AAG contains the same subsequence where all proline amino acids (24% of the total amino acid content) are mutated to alanine, maintaining the same codon adaptation index (Figure 1B).

SunTag-Renilla-RH1-MS2 is a plasmid containing the RH1 domain at the C-terminus of the Renilla open reading frame (Voigt et al., 2017) and it was used in the calibration experiment to image the mature proteins.

Cell line generation

$3 \cdot 10^5$ HeLa cells are seeded into a 6-well plate and the next day transfected with 2 µg of the FLPe recombinase plasmid (Addgene #20733) together with 2 µg of the plasmid carrying the SunTag reporter flanked by Flp-recombinase target sites. The next day, 5 µg/ml puromycin (Invivogen) is added to select for transfected cells. Two days later, the puromycin-containing medium is removed and cells are kept in growth medium containing 50 µM ganciclovir (Sigma-Aldrich) for 10 days to select for cells with successful RMCE. Single clones are isolated in a 96-well plate via FACS, based on cytoplasmic GFP intensity. Clones with moderate GFP background and responsive to doxycycline induction are selected for expansion.

Transient transfection

For the calibration experiment cells are transiently transfected with SunTag-Renilla-RH1-MS2 plasmid. Cells are seeded at $7.5 \cdot 10^3$ cells/well in a 8-well glass-bottom µ-slide (170 µm, ibidi) 48h before imaging. After 24h cells are transfected with Fugene transfection reagent according to the manufacturer's instructions with the SunTag plasmid but without the FLPe recombinase plasmid.

Generation of eIF5A knock-out cell line

Starting from the cell line stably expressing PPG, we generate an eIF5A knock out (KO) line using the single-guide RNA reported by Manjunath et al. (Manjunath et al., 2019) (eIF5A sgRNA #1 5' – AGAGGACCTTC GTCTCCCTG – 3') in an all-in-one Cas9-GFP plasmid. Single clones are selected with FACS sorting based on GFP intensity. Three clones are selected for further expansion, genotyping of the target site and western blotting of eIF5A and h-eIF5A. The clone with the strongest reduction in total eIF5A is chosen for imaging.

Live-cell imaging

Wild-type cells are seeded at a concentration of $7.5 \cdot 10^3$ cells/well in a 8-well glass-bottom µ-slide (170 µm, ibidi) 48h before imaging. eIF5A knock-out cells are seeded at $15 \cdot 10^3$ to reach a similar confluency at the acquisition. For the experiments involving GC7 treatment, 24h before imaging the growth medium is replaced with fresh medium supplemented with 1 mM AG in the control wells and with 1 µM GC7 and 1 mM AG in the treated wells. For experiments involving the eIF5A knockout, the growth medium is replaced with fresh medium (without AG). Before imaging the growth medium is replaced with fresh medium containing 1 µg/ml doxycycline (DOX) in order to induce the transcription of reporter mRNAs. After 40 minutes, the medium is further supplemented with Janelia Fluor 640 HaloTag ligand (JF646) (Promega), at 100 nM final concentration. After 20 minutes of incubation, the medium is removed and cells were washed in PBS. Cells are then kept in FluoroBrite DMEM (Thermo Fisher Scientific) supplemented with 10% FBS and 4 mM L-glutamine for 40 min before imaging and during the acquisition.

Cells are imaged using a spinning-disk confocal microscope (Visitron Spinning Disk CSU W1) equipped for live-cell imaging. Images are acquired using Visiview (Visitron) as single planes every 20 seconds and for 90 minutes, with 100 ms exposure time, 60X objective and hardware autofocus (see Table 1 for technical specifications). The focus is always set 0.8 µm away from the coverslip, yielding a focal plane close to the plasma membrane, ventral side of the cell. Because of the actin cortex, this area is usually rich in tethered reporter mRNAs. Cells are imaged sequentially in GFP and Cy5 channels with a single EMCCD camera.

In run-off assays harringtonine (HT) (Cayman Chemical) is pre-diluted in FluoroBrite and added to the cells (final concentration 3 µg/ml) mounted on the microscope stage. In practice, we add 53 µL of 20 µg/ µL working solution to 300 µL of FluoroBrite medium, pipetting up and down three times to ensure mixing. Image acquisition is started 30s – 60s after treatment (the delay between treatment and imaging is recorded for each acquisition). Time points are acquired every 20 s.

Laser power measurement and flat-field correction

488 nm and 640 nm laser power is measured at the beginning of each experiment using a power meter and a 10X Air objective (0.4 NA). During each acquisition – except in the calibration experiment – the percentage of laser power used is set to match 0.68 mW for the 488 nm laser and

Camera:	EMCCD Grayscale (ImagEMX2) 512 x 512 pixels, (16.0 x 16.5) µm/pixel 16-bit
Objective:	U PLAN S APO 60x 1.42NA
Laser lines:	488 nm 640 nm
Emission filters:	sdc GFP BP 525/50 (Chroma ET525/50m) sdc Cy5 BP 700/75 (Chroma ET700/75m)

Table 1. Technical specification of the live-cell imaging set-up.

2.5 mW for the 640 nm laser.

Every two months both microscope flat field and dark current are measured. For the flat-field measurement a uniformly fluorescent sample is prepared using ATTO 488 (AD 488-21, Atto-Tech) and ATTO 647 (AD 647-21, Atto-Tech), both in free acid form. Both dyes are dissolved in water at 1mM concentration (ATTO 647 dissolves inefficiently and the procedure can be improved by considering a different solvent) and then filtered with a 0.2 μm filter to remove aggregates. The solutions are added to 2% agarose gel to obtain a final concentration of 10 μM. The two fluorescent solutions are mixed 1:1 and a 40 μL drop is added to a microscopy slide and covered with a 22 x 22 mm coverslip, creating a thin, uniformly fluorescent, agarose layer. After drying, 20 xy positions are acquired in both channels with the 60X oil objective. The final flat-field image is obtained by averaging the 20 images and applying a median filter with radius 5 pixels. For the dark field measurement 100 time points are acquired without laser and without any specimen in both channels, and averaged to obtain the dark current image.

Mature protein intensity measurement

To measure the intensity of one mature protein (24 GFP molecules) we use a SunTag-Renilla-RH1MS2 reporter with an RH1 domain at the end of the coding sequence for actin-tethering of mature proteins (Voigt et al., 2017). The cells are seeded and prepared for imaging as previously described, except that DOX incubation is shortened to 20 min to avoid overcrowding of mature proteins in the cytoplasm. Prior to imaging, cells are treated with 100 µg/ml of puromycin (Sigma-Aldrich) to disassemble translating ribosomes, then imaged at 100% power (4.63 mW measured with a 10X Air objective) for 100 s at 10 Hz (Figure 1—figure Supplement 2A). Spot detection, tracking and intensity quantification (Figure 1—figure Supplement 2B) are performed in the GFP channel only, since mature proteins do not colocalize with the mRNA signal. The intensity of each spot is averaged over the 100 frames.

To build the intensity-power calibration curve, cells are treated with cycloheximide (Sigma-Aldrich) at 200 µg/ml final concentration 5 minutes before imaging, to block the ribosomes and have constant GFP intensity per mRNA molecule. Cells are imaged at 20%, 40%, 60%, 80% and 100% power, in rapid succession (10 frames per laser power at 0.5 Hz). This procedure revealed a linear relationship between the spot intensity and laser power (Figure 1—figure Supplement 2C), allowing us to infer the mature protein intensity at the laser power used for live imaging, from the value measured at 100% laser power. The final estimate (14 ± 2 a.u.) allows us to estimate the average number of ribosomes translating each report (Figure 1—figure Supplement 2D).

Quantification of Intensity Traces

Image processing

The live-cell images acquired during experiments are corrected for flat field and dark current as follows

$$C = \frac{O - DC}{FF - DC} \langle FF - DC \rangle \qquad (1)$$

where C is the corrected image, O is the original image, DC is the dark current image, FF is the flat field image and the average $\langle\ \rangle$ is calculated over all the pixels. Each experiment is corrected with the most recent dark current and flat-field images up to that date. Both the dark current and the flat-field measurements are very consistent across several months of imaging.

Spot tracking and intensity quantification

The spot detection is performed in the far red channel with TrackMate 7 (Fiji) (Ershov et al., 2022). Nuclei were excluded from detection because of the very high background due to JF646 nuclear localization. The tracking is performed using the TrackMate simple Linear Assignment Problem (LAP) tracker, with a maximal linking distance of 2 µm, a maximal distance for gap-closing of 3 µm and a maximal frame interval between two linked spots of 2 frames. The minimal track length is set to 5 minutes.

Given a 2 μm × 2 μm region around an mRNA, the intensity of the spots in the far red and green channel is inferred using Bayesian inference. The spot intensity in each channel is approximated as a 2D gaussian:

$$\mu_{2D}(x,y) = \frac{I}{2\pi w^2}\exp\left(-\frac{(x-x_0)^2}{2w^2} - \frac{(y-y_0)^2}{2w^2}\right) + g(x,y) \tag{2}$$

where I is the total spot intensity, w is the width of the spot, x_0 and y_0 are the coordinates of the spot center in micrometer units and $g(x, y)$ is a background given by

$$g(x,y) = c + b(y-y_0) + a(x-x_0), \tag{3}$$

with $a, b, c \in \mathbb{R}$. Hence, the background is modeled as a 2D plane with a and b representing the tilt in the x and y directions, respectively. This choice allowed us to model local changes in the cytoplasmic background, such as those happening when a spot is close to the cell membrane, and the GFP background is higher inside the cell than outside. Both a, b and c are parameters of the model. Finally, the intensity of a pixel is modeled as

$$i(x,y) \sim \mathrm{Normal}\left(\mu_{2D}(x,y), \sigma\right) \tag{4}$$

where σ is the standard deviation and it is estimated as the pixel standard deviation in the cytoplasmic background, and its value is fixed during the inference.

The priors on x_0 and y_0 are centered on the position x_r, y_r of the red spot, as given by TrackMate:

$$\begin{aligned} x_0 &\sim \mathrm{Normal}(x_r, 0.2) \\ y_0 &\sim \mathrm{Normal}(y_r, 0.2) \end{aligned} \tag{5}$$

with a standard deviation of 0.2 μm approximately corresponding to the average distance between the mRNA spot and the translation spot. The prior on the spot size is very peaked for regularization purposes

$$w \sim \mathrm{InvGamma}(\mu = 0.20, \sigma = 0.02). \tag{6}$$

The prior on the total intensity I is an “uninformative” prior

$$I \sim \mathrm{HalfNormal}(\sigma = 1000). \tag{7}$$

The background $g(x, y)$ is a 2D-plane, with

$$c \sim \mathrm{Normal}(\mu_c, \sigma_c) \tag{8}$$

where μ_c and σ_c are input parameters shared among all acquisitions, and they are estimated as the average and the standard deviation of the cytoplasm intensity across different cells and different experiments. Finally,

$$a, b \sim \mathrm{Normal}(0, 50) \tag{9}$$

which we include to account for the background variation when a spot is close to the transition between the cell interior and exterior. For the calibration experiment described in Mature protein intensity measurement, all the steps (detection, tracking and quantification) are done exclusively in the green channel, since mature proteins do not co-localize with the mRNA signal.

The inference is performed with the Hamiltonian Monte Carlo algorithm implemented in Stan (Team, 2024) through the PyStan Python interface.

Preprocessing and data filtering

The estimated spot intensity is affected by tracking errors and diffusion of the mRNA in the z direction of the focal plane. Cycloheximide (CHX) traces – acquired in the same imaging conditions as in Live-cell imaging – are used to estimate this noise and partially correct experimental traces. After regressing out the linear decay due to photobleaching, the standard deviation of the signal is calculated relative to the mean intensity of the trace; averaging over 236 traces yields a global estimate of the multiplicative noise (see Figure 2—figure Supplement 1 for examples of CHX traces and autocorrelation).

Next, all traces are smoothed with a low-pass filter (Butterworth filter), with a 1/120 Hz cut-off. At each time point the relative difference between the raw and smoothed trace is calculated; if this relative difference is higher than twice the multiplicative noise calculated from cycloheximide traces we replace it with the smoothed value. In practice, this corresponds to spikes or drops larger than ∼ 50% of the smoothed intensity.

For the run-off experiments the analysis is restricted to traces that start no later than 100 s after HT is added to the medium, so that translation is initially "close" to the distribution in the absence of inhibitor, considering that HT takes ∼ 60 s to enter the cells and be effective (Ingolia et al., 2009). Experiments with less than 10 traces in the control or in the perturbed (GC7/KO) conditions are excluded from the analysis. In addition, two acquisitions exhibited a delayed onset of translation, questioning the effectiveness of HT treatment in those instances, and were also excluded.

Mathematical models

Bayesian modeling of run-off traces

We consider a set of intensity traces $y_{r,t}$, where r is a trace index and t is a discrete time index, which is a multiple of the time step dt. We assume a common underlying stochastic process that can be described by an initiation rate α and an average elongation rate λ of the ribosomes along the transcript. In particular, we use a Hidden Markov Model (HMM) in which each trace $y_{r,t}$ is a noisy observation of a discrete Markov chain $N_{r,t}$, where $N_{r,t}$ is the number of translating ribosomes on the transcript r at time t. The inference is performed in Stan using a Markov chain Monte Carlo algorithm called Hamiltonian Monte Carlo (HMC). It uses an approximate Hamiltonian dynamics simulation based on numerical integration which is then corrected by performing a Metropolis acceptance step (Betancourt and Girolami, 2013).

In the following we describe how we derive the expressions of the initial state probability and the transition probabilities from a simple mean-field approximation of the ℓ-TASEP under the assumption of low ribosome density. At the end of the section we describe our assumptions related to the emission probability, i.e. the probability of observing the noisy signal y given the hidden state N.

Given a transcript of size L, we define the average ribosome density ρ, defined as the average fraction of transcript occupied by the ribosomes

$$\rho := \frac{\langle N \rangle \ell}{L}, \tag{10}$$

where ℓ is the length of transcript occupied by one ribosome and $\langle N \rangle$ is the average number of ribosomes on the transcript. The latter can be expressed as the product between the current J (number of particles per transcript site and per unit of time, equivalent to the protein synthesis rate) and the average residence time L/λ. Hence, Equation 10 can be written as

$$\rho = J\frac{\ell}{\lambda}. \tag{11}$$

In the initiation-limited regime as described in the continuum-limit approximation by Erdmann-Pham et al. (2020), the current is mainly determined by the initiation rate and it is given by

$$J = \alpha\frac{\lambda - \alpha}{\lambda + (\ell - 1)\alpha}. \tag{12}$$

In our model we assume that the initiation rate is low compared to elongation and ribosome excluded volume interactions can be neglected. In this case, we can expand the expression of J as a function of the ratio α/λ:

$$J = \alpha\left\{1 - \ell\frac{\alpha}{\lambda} + o\left[\left(\ell\frac{\alpha}{\lambda}\right)^2\right]\right\}, \tag{13}$$

to observe that when $\ell\frac{\alpha}{\lambda} \ll 1$, or equivalently $\lambda/\alpha \gg \ell$, we can approximate $J \approx \alpha$ and

$$\rho \approx \frac{\alpha}{\lambda}\ell \quad \text{if } \lambda/\alpha \gg \ell, \tag{14}$$

an approximation that we will use throughout our model. The initial state probability $\Theta_r(N)$ is the probability of observing N ribosomes at the beginning of trace r, given the model parameters. We set $t = 0$ when the run-off starts and we assume that this happens 60 s after HT is added to the medium (Ingolia et al., 2011). At $t = 0$ we can assume that the number of ribosomes N on the transcript is given by a Poisson distribution with mean $\alpha L/\lambda$, as in the absence of inhibitor, and where the low-density approximation in Equation 14 is assumed (Szavits-Nossan and Grima, 2023). Given that trace r starts at time $t_r \geq 0$, we assume that the ribosome distribution has not yet been substantially altered by initiation inhibition, and we approximate it by a Poisson distribution with mean corrected by t_r

$$p_r(N \mid \alpha, \lambda) = \text{Poisson}\left(N \,\middle|\, \alpha\left(\frac{L}{\lambda} - t_r\right)\right). \tag{15}$$

However, the probability of having $N = 0$ ribosomes on the transcript, hence the probability that a transcript is inactive at the time of HT treatment, may not follow the Poisson distribution – for instance, in the presence of switching between active and inactive states (whether or not the Poisson distribution is still a valid description for $N > 0$ will depend on the switching rates) (Szavits-Nossan and Grima, 2023). In practice, we describe the probability for a trace to be inactive as p_{off}, which is a parameter of the model and it is shared among all traces. Finally, the initial probability vector for trace r is given by

$$\Theta_r(N \mid \alpha, \lambda, p_{\text{off}}) = \begin{cases} p_{\text{off}} & N = 0 \\ p_r(N)/(1 - p_r(0)) & N > 0. \end{cases} \tag{16}$$

By choosing such a Poisson distribution, we assume that, in case of bursting, the durations of active and inactive periods are longer than the total mRNA translation time. For the inference we need to fix the maximum number of ribosomes $N_{\max}$ that can be translating the mRNA. We fix $N_{\max} = 100$, which corresponds to a density $\rho \approx 0.94$ for $L = 1066$ aa and $\ell = 10$ aa.

The transition probability $\Gamma(N \mid N', \alpha)$ is the probability of jumping from N' to N in a time dt, given the kinetic parameters α. Because at $t \geq 0$ initiation is inhibited, the only possible transitions are those satisfying $N \leq N'$, hence termination events.

When $\lambda/\alpha \gg \ell$ the waiting time distribution between successive termination events was derived in Szavits-Nossan and Grima (2023) and reads

$$P_{\text{term}}(t) = \alpha\frac{\lambda - \alpha}{\lambda - 2\alpha}\left(e^{-\alpha t} - e^{-(\lambda-\alpha)t}\right). \tag{17}$$

Since we are assuming $\alpha \ll \lambda/\ell < \lambda$ we can say that ${}^{\lambda-\alpha} \approx 1$ and $e^{-(\lambda-\alpha)t} \ll e^{-\alpha t}$. As a consequence, Equation 17 is approximately equal to the waiting time distribution between initiation events:

$$P_{\text{term}}(t) \approx \alpha e^{-\alpha t}. \tag{18}$$

So, we can define the transition probability from N to $N - k$ given α as

$$\Gamma(N - k \mid N, \alpha) = \text{Poisson}(k \mid \alpha dt). \tag{19}$$

For the inference, we need to set a maximum number of termination events $k_{\max}$ per time step dt. Considering that the initiation rates measured in the literature with similar systems range from less than 1 ribosome per min up to 5 ribosomes per min($\sim 0.08\ \text{s}^{-1}$) (Boersma et al., 2019; Livingston et al., 2023), we fixed $k_{\max} = 6$, since the probability of 6 initiation events in dt with $\alpha = 0.08\ \text{s}^{-1}$ is about 0.005.

We assume that the emission probability $\Phi(y \mid N, t)$ – the probability of observing an intensity y given N ribosomes at time t – is lognormal

$$\Phi(y \mid N, t) = \frac{1}{y\, s(t)\sqrt{2\pi}} \exp\left\{-\frac{[\ln y - \ln \mu(N, t)]^2}{2s^2(t)}\right\} \tag{20}$$

to account for different sources of noise arising during acquisition and not filtered by the intensity quantification and pre-processing steps (Wildner et al., 2023). These sources include z-diffusion of the mRNA, irregular background, fluctuations in laser intensity, photon shot noise, etc. In addition, this choice aims to effectively describe the noise deriving from the uncertainty regarding the ribosome distribution along the mRNA. For a time t and N ribosomes, an equivalent representation in terms of random variables is

$$Y = \mu \exp(sX) \tag{21}$$

where X is a standard normal random variable, $\mu \coloneqq \mu(N, t)$ and $s \coloneqq s(t)$. In practice, the measured spot intensity y is given by the true signal μ multiplied by an exponentially-transformed gaussian noise.

The following discussion is dedicated to the derivation of $\mu(N, t)$ and $s(t)$ which explains the explicit dependence of the emission probability on time. Notice that the noise parameter s can be expressed in terms of the expectation $\langle y(N, t) \rangle$ and variance $\sigma_y^2(N,t)$ of y as

$$s^2(N,t) = \ln\left(1 + \frac{\sigma_y^2(N,t)}{\langle y(N,t)\rangle^2}\right). \tag{22}$$

The denoised signal μ is a function of the number of translating ribosomes N and time t and, since we are neglecting the precise ribosome positions along the mRNA, it is an average over all the possible configurations $C_{N,t}$ of N ribosomes along the mRNA at time t of run-off. In practice, we consider that the ribosomes are uniformly distributed along the mRNA. We express it as

$$\mu(N,t) = \left\langle I(C_{N,t}) \right\rangle_{C_{N,t}} = b_0 + \gamma(t) N i_{\mathrm{MP}}, \tag{23}$$

where $I(C_{N,t})$ is the intensity generated by the ribosome configuration $C_{N,t}$, b_0 is the offset, i_{MP} is the intensity of one mature protein and γ is a time-dependent correction factor that accounts for the fact that ribosomes translating the SunTag have intensity smaller than i_{MP}. It depends on time because during run-off the ensemble of possible ribosomes configurations along the mRNA changes, as ribosomes progress on the transcript. Given the length of the SunTag L_S, we can express $\gamma(t)$ in terms of the elongation rate λ (see Mean intensity correction factor during run-off for the full derivation):

$$\gamma(t) = 1 - \frac{(L_S - \lambda t)^2}{2L_S(L - \lambda t)} \quad \text{for } \lambda t \le L_S, \tag{24}$$

which agrees with the result for $t = 0$ previously reported (Aguilera et al., 2019). When all ribosomes have completed the translation of the SunTag, and approximately for $\lambda t > L_S$, $\gamma(t) = 1$. The value of γ over time agrees well with simulation results over a broad range of ribosome densities (Figure 2C and Figure 2—figure Supplement 2A).

We can compute the variance of the total spot intensity in a similar way

$$\sigma^2(N,t) := \mathrm{Var}\left[I(C_{N,t})\right]_{C_{N,t}} = v(t) N i_{\mathrm{MP}}^2 \tag{25}$$

where v is given by (see Intensity variance for the derivation)

$$v(t) = \frac{1}{L - \lambda t}\left(L - \frac{2L_S}{3} - \frac{(\lambda t)^3}{3L_S^2}\right) - \left(1 - \frac{(L_S - \lambda t)^2}{2L_S(L - \lambda t)}\right)^2. \tag{26}$$

The agreement with simulations is good at low densities, while it significantly deteriorates at higher densities (Figure 2D and Figure 2—figure Supplement 2A). This is expected as the expressions of both $\gamma(t)$ and $v(t)$ are derived under the assumption that the ribosome density is sufficiently low that excluded-volume interactions can be neglected, and the probability to find a ribosome at position i can be approximated by N/L.

We can write the total variance $\sigma_y^2(N,t)$ of the signal $y(N, t)$ as the sum of the variance σ^2_{meas} due to measurement noise and variance $\sigma_C^2(N,t)$ due to the ribosome configurations on the mRNA, and effectively subsume both of them under the log-normal noise. So, the observed intensity y is a random variable because of the unknown measurement process and because of the unknown underlying ribosome distribution. Then, the normalized variance in the expression of s (Equation 22) reads

$$\frac{\sigma_y^2(N,t)}{\langle\, y(N,t)\,\rangle^2} = \frac{\sigma^2_{\mathrm{meas}}(N,t)}{\langle\, y(N,t)\,\rangle^2} + \frac{\sigma_C(N,t)^2}{\langle\, y(N,t)\,\rangle^2} \tag{27}$$

where the average is intended over the measurements and the configurations $C_{N,t}$. We approximate the first term in the sum as

$$\frac{\sigma^2_{\mathrm{meas}}(N,t)}{\langle\, y(N,t)\,\rangle^2} \approx \frac{\sigma^2_{\mathrm{meas}}(N,t)}{\langle\, I(C_{N,t})\,\rangle^2_{\mathrm{meas}}} \tag{28}$$

where the average on the right-hand side is performed over the measurements and where we assume that the right-hand side is a constant multiplicative noise that does not depend on the specific ribosome configuration, nor the intensity, and subsumes the uncertainty in the acquisition process. We approximate the second term in the sum as

$$\frac{\sigma_C^2(N,t)}{\langle\, y(N,t)\,\rangle^2} \approx \frac{\sigma^2(N,t)}{\langle\, I(C_{N,t})\,\rangle^2_{C_{N,t}}} = \frac{\sigma^2(N,t)}{\mu^2} = \frac{1}{N}\frac{\nu(t)}{\gamma^2(t)} \tag{29}$$

and, by substituting N with the average $\langle\, N\,\rangle = \alpha L/\lambda$,

$$\frac{\sigma_C^2(N,t)}{\langle\, y(N,t)\,\rangle^2} \approx \frac{\lambda}{\alpha L}\frac{\nu(t)}{\gamma^2(t)}. \tag{30}$$

Finally, we can use Equation 22 to write the final estimate for s

$$s(t) = \sqrt{\ln\left(1 + \frac{\sigma^2_{\mathrm{meas}}}{\langle I\,\rangle^2_{\mathrm{meas}}} + \frac{\lambda}{\alpha L}\frac{\nu(t)}{\gamma^2(t)}\right)}, \tag{31}$$

where we have omitted the dependence of I on ribosome configurations $C_{N,t}$ for simplicity. Notice that $s(t)$ depends solely on time and on the kinetic parameters α and λ. When there are no ribosomes on the transcript or when $\nu(t) = 0$ (all ribosomes have terminated the translation of the SunTag) the noise is simply given by

$$s_0 := \sqrt{\ln\left(1 + \frac{\sigma^2_{\mathrm{meas}}}{\langle I\,\rangle^2_{\mathrm{meas}}}\right)}. \tag{32}$$

Finally, the emission probability is given by

$$\Phi(y\,|\,N,t) = \frac{1}{ys(t)\sqrt{2\pi}}\exp\left\{-\frac{[\ln y - \ln\mu(N,t)]^2}{2s^2(t)}\right\} \tag{33}$$

where $\mu(N,\, t)$ is given by Equation 23 and $s(t)$ is given by Equation 31.

For each experiment we estimate the traces baseline b_0 in Equation 23 and the term $\sigma^2_{\mathrm{meas}}/\langle I\,\rangle^2$ in Equation 31 independently of the other model parameters. The offset b_0 is estimated as the average intensity of the untranslated traces: because it is very consistent across experiments and conditions we fix it to its average value across experiments, $b_0 = 6$ a.u.

The term $\sigma^2_{\mathrm{meas}}/\langle I\,\rangle^2$ determines the noise parameter s_0 (Equation 32), which accounts for the measurement noise alone (and not for the uncertainty of the ribosome distribution). It is estimated independently for each experiment (all conditions together) by calculating the variance of untranslated traces σ^2_{meas} divided by the mean intensity squared of the trace $\langle\, I\,\rangle^2$. The values obtained in this way correspond to s_0 ranging from 0.31 to 0.38, with mean 0.34 and standard deviation 0.02 – across experiments. In an independent estimate from raw CHX traces described in Preprocessing and data filtering, we obtain $s_0 = 0.36$, a value comparable to what was obtained for untranslated traces (Figure 2—figure Supplement 1A, B).

Mean intensity correction factor at $t = 0$

We compute here the correction factor γ for the total intensity of N ribosomes used in Bayesian modeling of run-off traces, at $t = 0$. The number of epitopes n on the SunTag as a function of the position x can be approximated as

$$n(x) = \frac{n_S}{L_S - 1}(x-1) \quad x \le L_S \tag{34}$$

where $n_S = 24$ is the total number of epitopes encoded in the SunTag and L_S is the SunTag length. For $x > L_S$ we have $n(x) = n_S$. In the following we will consider the intensity normalized by i_{MP} and use the fraction of epitopes $f(x) := n(x)/n_S$. We can compute the average intensity for $N = 1$ ribosomes translating the SunTag as

$$\mu_1 = \sum_{x=1}^{L} f(x)p(x) \tag{35}$$

where $p(x)$ is the probability of finding a ribosome at x. After approximating $p(x) = 1/L$ and using the definition of $f(x)$ we have

$$\mu_1 = \frac{1}{L}\sum_{x=1}^{L} f(x) = \frac{1}{L}\left[\frac{1}{L_S-1}\sum_{x=1}^{L_S}(x-1) + (L - L_S)\right]$$

which yields

$$\mu_1 = 1 - \frac{L_s}{2L} = \gamma, \tag{36}$$

where we used

$$\sum_{x=1}^{L_S}(x-1) = \frac{L_S(L_S-1)}{2}.$$

For $N > 1$ we have $\mu_N = N\mu_1 = N\gamma$.

Mean intensity correction factor during run-off

We compute here the correction factor γ for the total intensity of N ribosomes used in Bayesian modeling of run-off traces during run-off. While ribosomes run off the transcripts, the proportion of ribosomes on the SunTag changes, which can be taken into account by a time-dependent correction factor $\gamma(t)$, dependent on the elongation rate λ (that we assume to be uniform along the transcript). We can approximate the occupation probability as

$$p(x \mid N, t) = \begin{cases} 0 & \forall x \le x_t \\ N/(L - x_t) & \forall x > x_t \end{cases} \tag{37}$$

where we defined $x_t := \lfloor \lambda t \rfloor \in [1, L]$, the average distance covered by the ribosomes in a time t. With the same algebra as before we get $\mu_N = N\gamma(t)$ with

$$\gamma(t) = \frac{1}{L - x_t}\left(L - \frac{L_S}{2} - \frac{x_t(x_t-1)}{2(L_S-1)}\right) \simeq 1 - \frac{(L_S - \lambda t)^2}{2L_S(L - \lambda t)} \quad \forall t \le L_S/\lambda \tag{38}$$

else $\gamma(t) = 1$.

Intensity variance

We compute the variance $V_1(y)$ of the intensity y produced by $N = 1$ ribosomes at time t during run-off, in units of i^2_{MP} corresponds to when run-off starts). The variance for N ribosomes will be the sum of the variances: $V_N(y) = NV_1(y)$. We can write the variance as

$$V_1(y) = \langle y^2 \rangle - \mu_1^2 \tag{39}$$

with μ_1 given by Equation 36 and

$$\langle y^2 \rangle = \sum_{x=x_t}^{L} f(x)^2 p(x) = \frac{1}{L - x_t}\left(\sum_{x=1}^{L} f(x)^2 - \sum_{x=1}^{x_t} f(x)^2\right) \tag{40}$$

where $x_t := \lfloor \lambda t \rfloor \in [1, L]$ as before. The first summation gives

$$\begin{aligned}\sum_{x=1}^{L} f(x)^2 &= \frac{1}{(L_S-1)^2}\sum_{x=1}^{L_S}(x-1)^2 + L - L_S\\ &= \frac{1}{(L_S-1)^2}\left(\frac{L_S(L_S+1)(2L_S+1)}{6} - L_S(L_S+1) + L_S\right) + L - L_S\\ &= \frac{L_S(2L_S-1)}{6(L_S-1)} + L - L_S \approx L - \frac{2L_S}{3}.\end{aligned}$$

The second term

$$\begin{aligned}\sum_{x=1}^{x_t} f(x)^2 &= \frac{1}{(L_S-1)^2}\sum_{x=1}^{x_t}(x-1)^2\\ &= \frac{1}{(L_S-1)^2}\left(\frac{x_t(x_t+1)(2x_t+1)}{6} - x_t(x_t+1) + x_t\right)\\ &= \frac{x_t(2x_t-1)(x_t-1)}{6(L_S-1)^2} \approx \frac{(\lambda t)^3}{3L_S^2}.\end{aligned}$$

Finally,

$$\langle y^2 \rangle \approx \frac{1}{L-\lambda t}\left(L - \frac{2L_S}{3} - \frac{(\lambda t)^3}{3L_S^2}\right).$$

The final results is then

$$V_1(y) = \frac{1}{L-\lambda t}\left(L - \frac{2L_S}{3} - \frac{(\lambda t)^3}{3L_S^2}\right) - \left(1 - \frac{(L_S-\lambda t)^2}{2L_S(L-\lambda t)}\right)^2. \tag{41}$$

Simulation of SunTag traces

SunTag traces are obtained *in silico* by simulating the homogeneous ℓ-TASEP with the Gillespie algorithm (Gillespie, 1976). The algorithm is formulated for a general set of site-specific elongation rates λ_i, but for the purposes of this study we simulate a homogeneous system with all elongation rates equal to λ; for simplicity, the termination rate λ_{L-1} is considered equal to the elongation rate. We first describe the algorithm used for the ℓ-TASEP simulations without bursting, and then we describe its generalization for a bursting initiation.

The simulation starts at $t = 0$. At any step of the simulation, a vector $\tau = (\tau_0, \ldots, \tau_{L-1}) \in \{0, 1\}^L$ specifies the ribosome occupancy state of the mRNA (of length L) and a list Ω specifies all the possible reactions and their associated time. In particular, $\omega_k \in \Omega$ is a list of two elements: $\omega_k[0] \in \{0, \ldots, L-1\}$ specifies the event and $\omega_k[1] \in \mathbb{R}^+$ its time of occurrence. $\omega_k[0] = -1$ corresponds to an initiation event, $\omega_k[0] = L-1$ a termination event, and $\omega_k[0] \in \{0, \ldots, L-2\}$ an elongation event at position $\omega_k[0]$. The list Ω is kept ordered according to the time of occurrence, such that the first element of the list is the one occurring earlier in time and $\omega_k[1] \leq \omega_{k+1}[1]$. In the following $\mathrm{Exp}(\lambda)$ is the exponential distribution with scale parameter λ.

1. Remove the event ω_1 from Ω (let $i = \omega_1[0]$ and $dt = \omega_1[1]$).
2. Update the total time $t \leftarrow t + dt$ and the single events time $\omega_k[1] \leftarrow \omega_k[1] - dt\ \forall k$.
3. Update the occupancy state:
   a. If $i = -1$ (initiation), then set $\tau_0 = 1$.
   b. If $i = L-1$ (termination), then set $\tau_{L-1} = 0$.
   c. If $i \in \{0, \ldots, L-2\}$ (elongation), then set $(\tau_i, \tau_{i+1}) \leftarrow (0, 1)$.
4. Find possible new reactions and sample their associated times:
   a. If $i = L-1$ and $\tau_{L-\ell-1} = 1$, then sample $t_0 \sim \mathrm{Exp}(\lambda_{L-\ell-1})$ and insert $(L-\ell-1, t_0)$ into Ω.
   b. If $i \in \{0, \ldots, L-2\}$
      i. If $i = \ell - 1$, then sample $t_0 \sim \mathrm{Exp}(\alpha)$ and insert $(-1, t_0)$ into Ω.
      ii. If $i > \ell - 1$ and $\tau_{i-\ell} = 1$, then sample $t_0 \sim \mathrm{Exp}(\lambda_{i-\ell})$ and insert $(i-\ell, t_0)$ into Ω.
      iii. If $i < L-\ell$ and $\tau_{i+\ell+1} = 0$ or if $L-\ell \leq i \leq L-2$, then sample $t_0 \sim \mathrm{Exp}(\lambda_{i+1})$ and insert $(i+1, t_0)$ into Ω.
5. Go to 1.

The algorithm easily generalizes to a 2-state telegraph model for initiation: the state of the system is now described by τ and $G \in \{0, 1\}$, with $G = 0$ if the transcript is inactive and $G = 1$ if the transcript is active. In the inactive state initiation is not possible, while in the active state initiation is an exponential process with rate α. In practice, we add the reaction $\omega_k[0] = -2$ that corresponds to a change in transcript activity, $G \rightarrow \neg G$. The reaction time is drawn from an exponential distribution

$$\omega_k[1] \sim \begin{cases} \mathrm{Exp}(k_1) & \text{if } G = 0 \\ \mathrm{Exp}(k_0) & \text{if } G = 1. \end{cases}$$

We first simulate the model for 10^3 iterations of burn-in; for each trace we record the state of the system every $dt = 20$ s for a total duration of 90 min. The final state is used as a starting point to simulate the dynamics for 10^3 iterations before recording the following trace. For the run-off, we remove the initiation events possibly stored in Ω after the initial burn-in and before simulating the first trace with $\alpha = 0$. The last state of burn-in is then used as a starting point to simulate the dynamics for 10^3 more iterations before recording the following run-off.

To obtain the translation signal, we compute the number of epitopes $n(t)$ synthesized at each time point t by all the translating ribosomes. This is given by the scalar product

$$n(t) = \boldsymbol{w} \cdot \boldsymbol{\tau}(t) \tag{42}$$

where $\boldsymbol{w} \in \mathbb{N}^L$, with L the transcript size, containing the cumulative number of epitopes along the transcript, and $\tau \in \{0, 1\}^L$ is the time-dependent occupation vector, with

$$\tau_i = \begin{cases} 0 & \text{if site } i \text{ is occupied} \\ 1 & \text{if site } i \text{ is free.} \end{cases} \tag{43}$$

The total intensity is given by $I_S(t) = \frac{i_{MP}}{24} n(t)$, where i_{MP} is the intensity of one mature protein and the suffix S stands for “simulated”.

We add an offset b_0 to the intensity and we apply a log-normal noise to obtain the final simulated, noisy signal $y_S(t)$:

$$P(y_S(t) \mid \mu_S(t), s_0) = \frac{1}{y_S(t) s_0 \sqrt{2\pi}} \exp\left\{ -\frac{(\ln y_S(t) - \ln \mu_S(t))^2}{2 s_0^2} \right\} \tag{44}$$

where $\mu_S(t) = b_0 + I_S(t)$ and s_0 is the measurement noise, as described at the end of Bayesian modeling of run-off traces.

The duration of each trace is drawn from an exponential distribution with mean 10 min to mimic the tracking duration of one mRNA molecule in experiments. Simulated run-off traces are longer than 5 min (as in experiments, see Spot tracking and intensity quantification) and start at $t = 0$, when the run-off starts (approximately 60 s after HT addition, see Preprocessing and data filtering).

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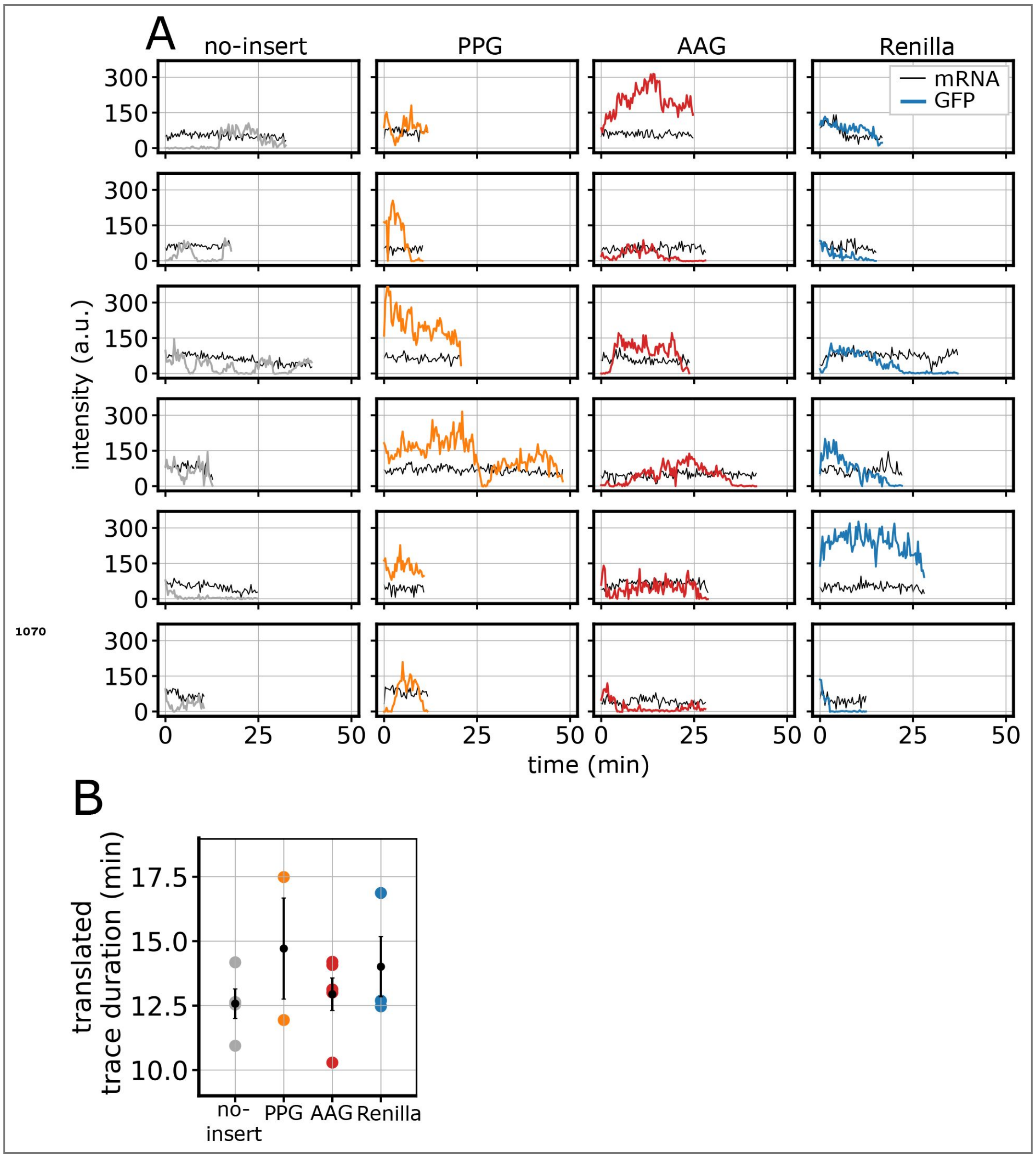


Figure 1—figure supplement 1. SunTag traces in control conditions. (A) Representative SunTag traces for different reporters in control conditions. **(B)** Duration of translated traces in control conditions (in black, average between experiments ± SEM).

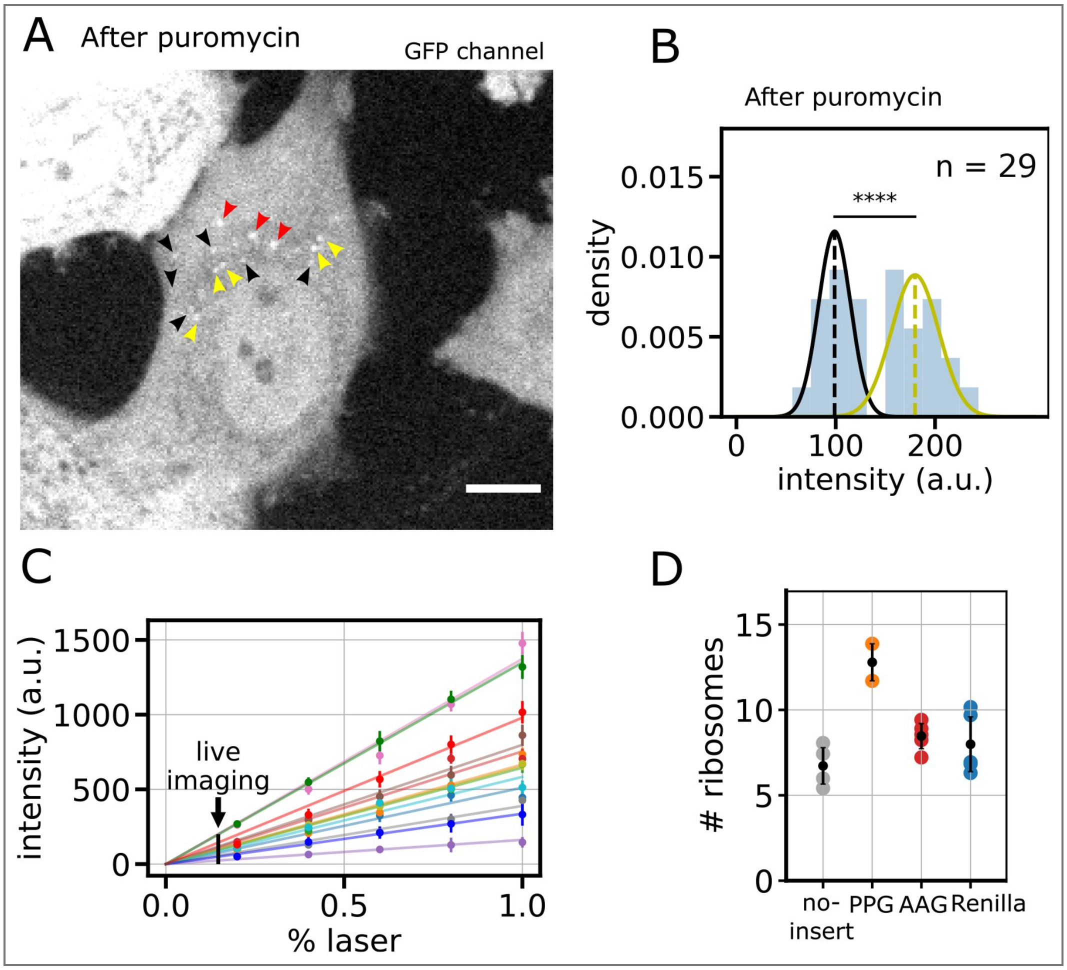


Figure —figure supplement 2. Experimental measurement of mature protein intensities to estimate number of translating ribosomes.

A) Diffraction-limited spots in puromycin-treated cell expressing SunTag-Renilla-RH1-MS2, GFP channel. SunTag mature proteins are tethered to actin via the RH1 domain, that prevents their free diffusion in the cytoplasm, allowing to measure their intensity. Because of the low intensity, cells were imaged at 100% laser power. Black, yellow and red arrows indicate spots of different sizes and intensities. The image is part of a larger field-of-view, acquired with a 60X oil-objective. Scale bar: 10 µm. **B)** Histogram showing the average intensity distribution of 29 spots of lower intensity, after puromycin treatment, at 100% laser power. Black and yellow lines represent gaussian fits of a lower intensity (black) and a higher intensity (yellow) spot population. **C)** GFP intensity at increasing laser power of CHX-treated SunTag reporters. After CHX treatment, the intensity of the GFP signal is constant because translating ribosomes are blocked on the mRNA. This allows to estimate the dependence of measured intensity on laser power, which results to be linear. Each color represents one CHX-treated mRNA imaged at increasing laser power. **(D)** Number of ribosomes per translated period in traces longer than 20 min (in black, average across experiments ± SEM), as determined from the measurement of one mature protein intensity (A–C).

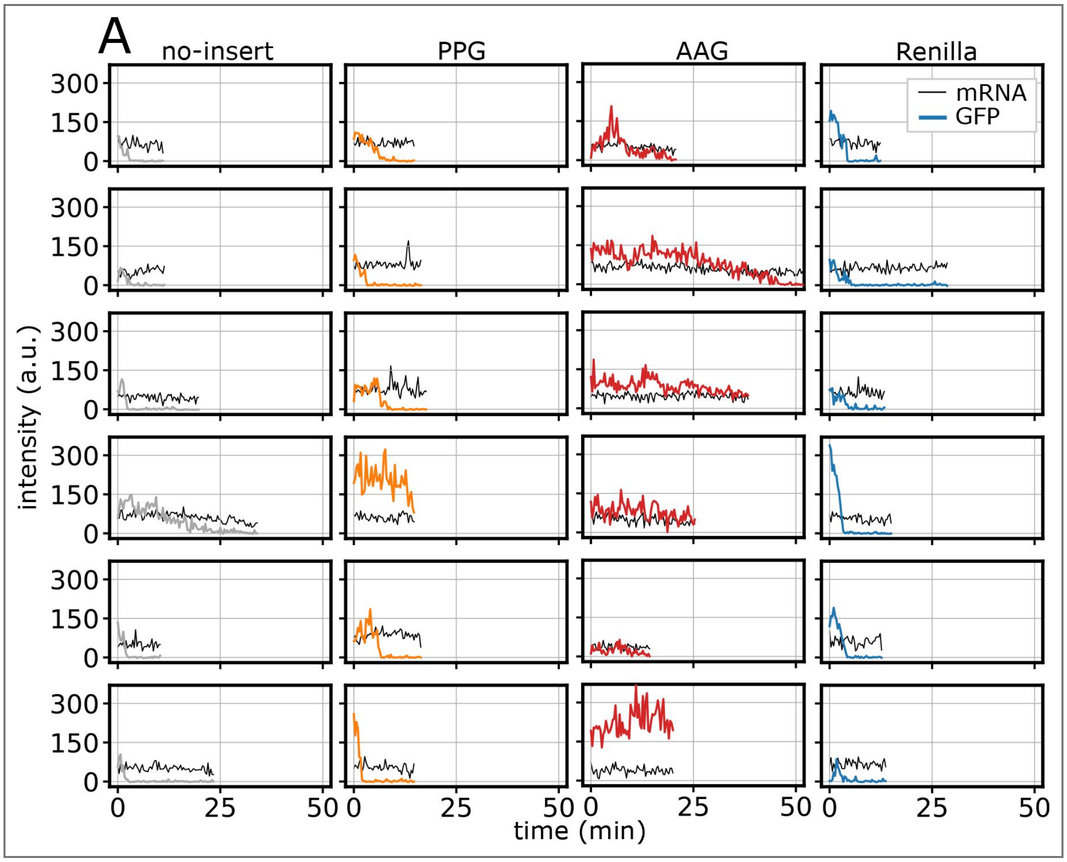


Figure 2—figure supplement 1. **SunTag run-off traces in control conditions after HT addition.**

(A) Representative SunTag traces for different reporters in control conditions after HT addition.

Figure 2—figure supplement 2. TASEP-based inference accurately estimates ribosome density on simulated data despite correlations between α and i_{MP}.

A) Agreement of $\gamma(t)$ (top) and $\nu(t)$ (bottom) with simulations for different values of λ, with $\alpha = 1/30s^{-1}$. While λ decreases (left to right) the average ribosome density ρ increases and the agreement between $\nu(t)$ and simulations significantly deteriorates. **B)** Correlation between the relative error on α and the relative error on i_{MP}, for L = 1066aa and L = 714aa and s_0 = 0.4. Black and red dots represents data with $\rho > 0.10$ and $\rho \leq 0.10$, respectively. The Pearson correlation coefficient r and the p-value p is indicated for all data (n = 12 simulations) and the black line shows the linear fit. The initiation rate α and on the mature protein intensity i_{MP} are negatively correlated, but at low densities ($\rho < 0.10$) the mature protein intensity is inferred more accurately and the correlation is less important. Simulation parameters: $\lambda \in \{0.5, 1, 3, 5\}$ aa/s, $\alpha \in \{1/120, 1/60, 1/30\}$ s^{-1}, i_{MP} = 10 a.u. **C)** Inferred average density $\rho = \alpha L/\lambda$ at the beginning of run-off *vs* true average density as measured in simulations (s_0 = 0.4). The black line shows the $y = x$ line. In all panels error bars represent 95%-confidence intervals.

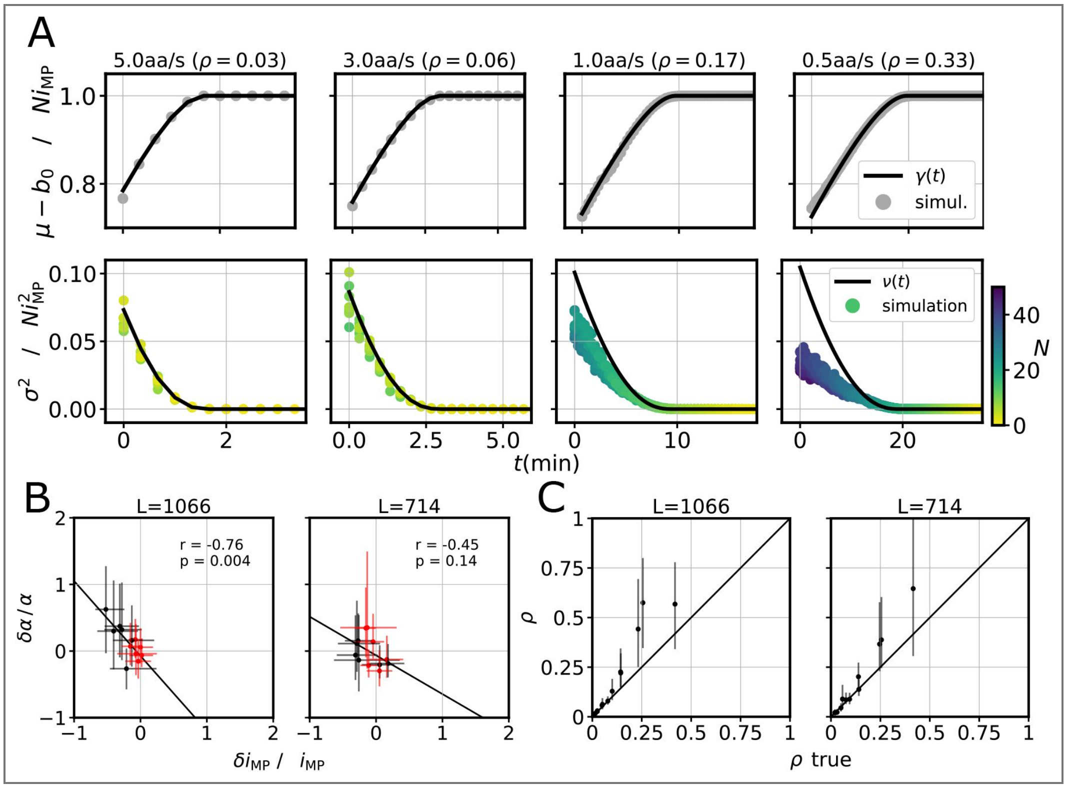
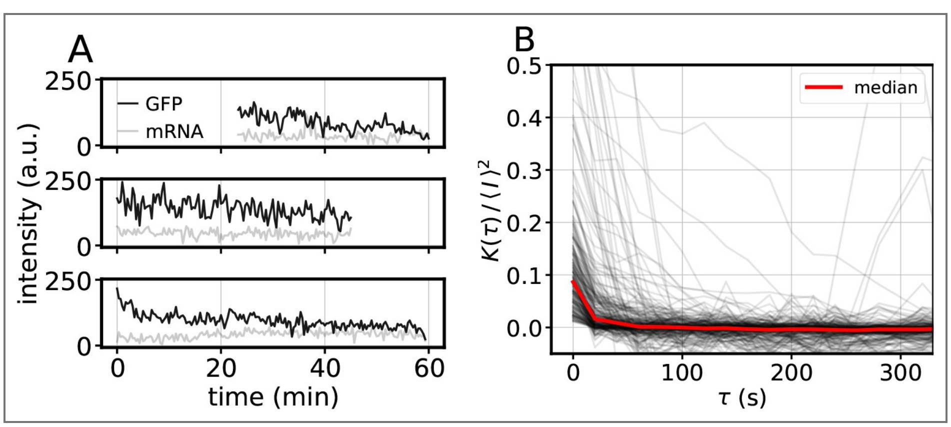


Figure 3-Figure supplement 1. Global measurement noise estimation from CHX traces.

A) Examples of GFP intensity traces after CHX treatment (black) and corresponding mRNA signal (grey). B) Autocorrelation ***K(τ)*** normalized by the square of the temporal mean intensity ***⟨ I ⟩*** for each CHX-treated trace (black lines) and median over all traces (red line), for different time delays τ. ***K(0)/ ⟨ I ⟩*** averaged over 236 traces yields an estimate of the multiplicative noise factor.



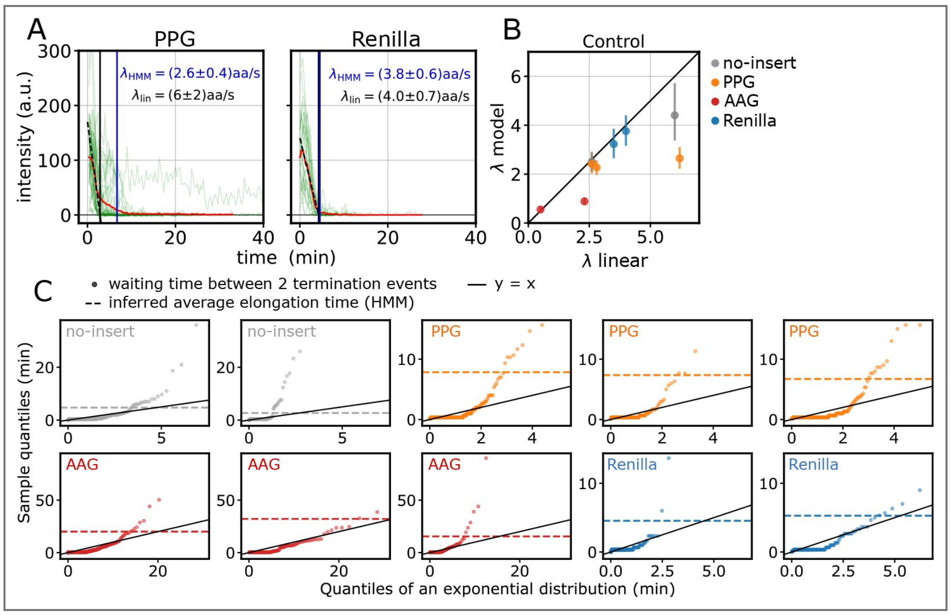


Figure 3—figure supplement 2. **HMM approach predicts lower elongation speed than simple linear regression and suggests the presence of bursting and stalling.**

(A) Harringtonine runoff traces of PPG and Renilla (2 acquisitions). The black dashed line represents a linear regression of the average intensity decay (red). Vertical lines indicates the total elongation time as predicted by the linear fit (black) or by the HMM (blue). **(B)** Elongation rate predicted by the HMM *vs* elongation rate predicted by the linear-regression approach. **(C)** Q-Q plot of the waiting times between termination events, compared to an exponential distribution with mean the inverse initiation rate (as inferred by the HMM). Each plot represents one acquisition, and each dot the time between two termination events. The dashed line represents the total elongation time as inferred by the HMM model, and the full line the bisector. Waiting times lying above the dashed line are likely to be strong stalling events.

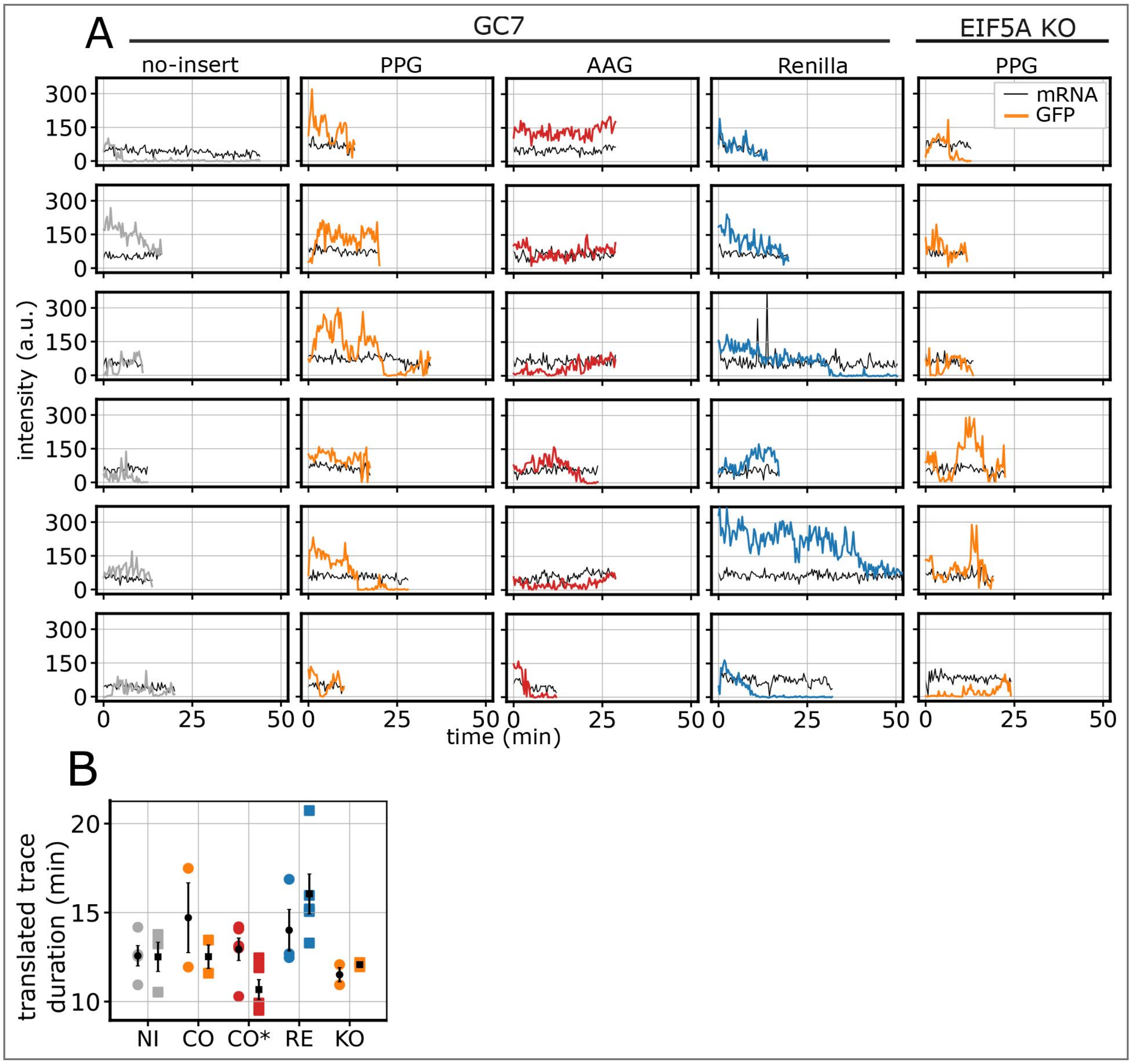


Figure 4—figure supplement 1. **SunTag traces in perturbed conditions.**

(A) Representative SunTag traces for different reporters in perturbed (GC7/EIF5A KO) conditions. **(B)** Duration of translated traces in perturbed conditions (average across experiments ± SEM).

Figure 5—figure supplement 1. SunTag run-off traces in perturbed conditions after HT addition.

(**A**) Representative SunTag traces for different reporters in perturbed conditions (GC7/EIF5A KO) after HT addition.

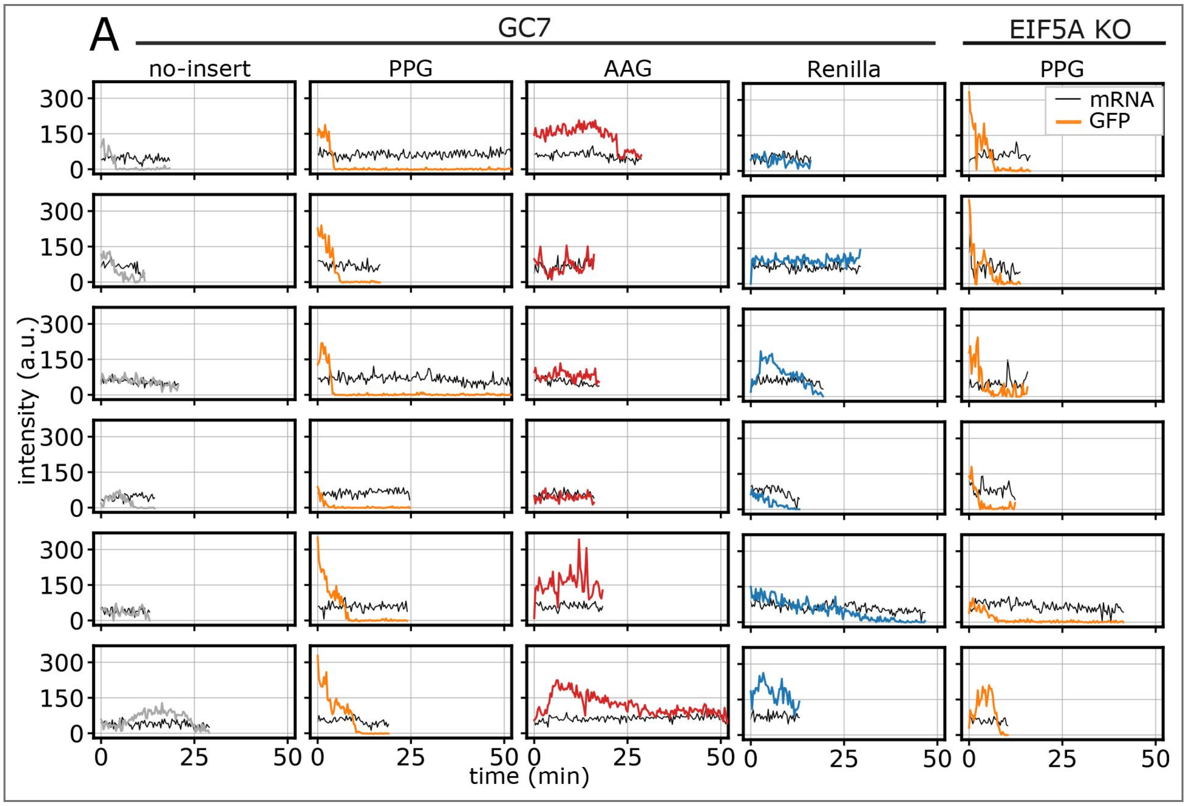
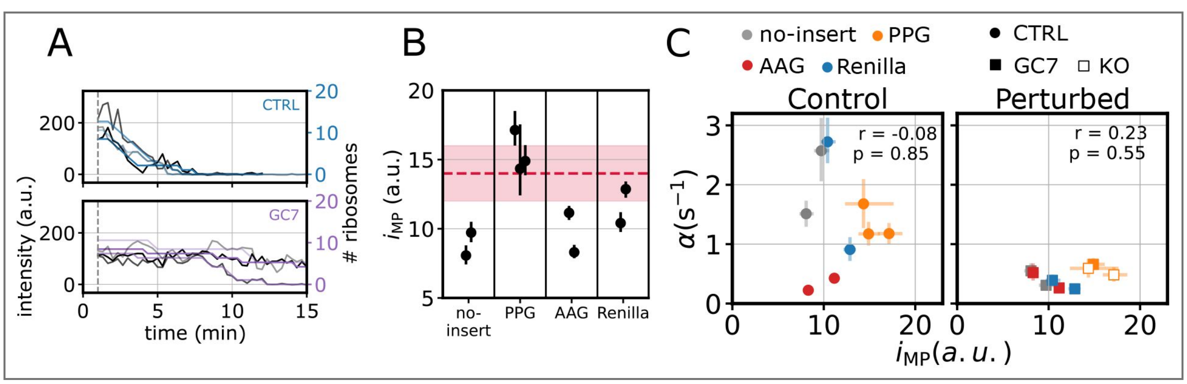


Figure 5—figure supplement 2. Inferred mature protein intensities are in good agreement with the experimental measurement.

A) Representative PPG traces in control and GC7-treated conditions, together with the inferred number of ribosomes. **B)** Mature protein intensity i_{MP} for each experiment, subdivided by reporter. The red dashed line indicates the mature protein intensity as measured by an independent calibration experiment and the filled area the uncertainty on the measurement. **C)** Initiation rate *vs* mature protein intensity, for both control and perturbed conditions (*n* = 10 experiments): α initiation rate and i_{MP} mature protein intensity.



Data availability

Our computational models for analysis of SunTag traces are available on Github (*https://github.com/naef-lab/suntag-analysis*). SunTag run-off and steady-state raw images are available on Zenodo (*https://doi.org/10.5281/zenodo.17669331*).

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Additional files

Fig 1-video 1. Time-lapse imaging of HeLa cells expressing the AAG reporter.

Fig 2-video 2. Time-lapse imaging of HeLa cells expressing the PPG reporter after harringtonine treatment.

Fig 2-video 2. Time-lapse imaging of HeLa cells expressing the AAG reporter after harringtonine treatment.

Fig 4-video 1. Time-lapse imaging of HeLa cells expressing the PPG reporter after 24h GC7 treatment.

Fig 4-video 2. Time-lapse imaging of EIF5A KO HeLa cells expressing the PPG reporter.

Fig 5-video 1. Time-lapse imaging of HeLa cells expressing the PPG reporter after 24h GC7 treatment and harringtonine addition.

Fig 5-video 2. Time-lapse imaging of HeLa cells expressing the AAG reporter after after 24h GC7 treatment and harringtonine addition.

Fig 5-video 3. Time-lapse imaging of EIF5A KO HeLa cells expressing the PPG reporter after harringtonine addition.

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Peer reviews

Reviewer #1 (Public review):

Summary:

In this study, Lamberti et al. investigate how translation initiation and elongation are coordinated at the single-mRNA level in mammalian cells. The authors aim to uncover whether and how cells dynamically adjust initiation rates in response to elongation dynamics, with the overarching goal of understanding how translational homeostasis is maintained. To this end, the study combines single-molecule live-cell imaging using the SunTag system with a kinetic modeling framework grounded in the Totally Asymmetric Simple Exclusion Process (TASEP). By applying this approach to custom reporter constructs with different coding sequences, and under perturbations of the initiation/elongation factor eIF5A, the authors infer initiation and elongation rates from individual mRNAs and examine how these rates covary.

The central finding is that initiation and elongation rates are strongly correlated across a range of coding sequences, resulting in consistently low ribosome density ({less than or equal to}12% of the coding sequence occupied). This coupling is preserved under partial pharmacological inhibition of eIF5A, which slows elongation but is matched by a proportional decrease in initiation, thereby maintaining ribosome density. However, a complete genetic knockout of eIF5A disrupts this coordination, leading to reduced ribosome density, potentially due to changes in ribosome stalling resolution or degradation.

Strengths:

A key strength of this work is its methodological innovation. The authors develop and validate a TASEP-based Hidden Markov Model (HMM) to infer translation kinetics at single-mRNA resolution. This approach provides a substantial advance over previous population-level or averaged models and enables dynamic reconstruction of ribosome behavior from experimental traces. The model is carefully benchmarked against simulated data and appropriately applied. The experimental design is also strong. The authors construct matched SunTag reporters differing only in codon composition in a defined region of the coding sequence, allowing them to isolate the effects of elongation-related features while controlling for other regulatory elements. The use of both pharmacological and genetic perturbations of eIF5A adds robustness and depth to the biological conclusions. The results are compelling: across all constructs and conditions, ribosome density remains low, and initiation and elongation appear tightly coordinated, suggesting an intrinsic feedback mechanism in translational regulation. These findings challenge the classical view of translation initiation as the sole rate-limiting step and provide new insights into how cells may dynamically maintain translation efficiency and avoid ribosome collisions.

Assessment of Goals and Conclusions:

The authors successfully achieve their stated aims: they quantify translation initiation and elongation at the single-mRNA level and show that these processes are dynamically coupled to maintain low ribosome density. The modeling framework is well suited to this task, and the conclusions are supported by multiple lines of evidence, including inferred kinetic parameters, independent ribosome counts, and consistent behavior under perturbation.

Impact and Utility:

This work makes a significant conceptual and technical contribution to the field of translation biology. The modeling framework developed here opens the door to more detailed and quantitative studies of ribosome dynamics on single mRNAs and could be adapted to other imaging systems or perturbations. The discovery of initiation-elongation coupling as a general feature of translation in mammalian cells will likely influence how researchers think about translational regulation under homeostatic and stress conditions.

The data, models, and tools developed in this study will be of broad utility to the community, particularly for researchers studying translation dynamics, ribosome behavior, or the effects of codon usage and mRNA structure on protein synthesis.

Context and Interpretation:

This study contributes to a growing body of evidence that translation is not merely controlled at initiation but involves feedback between elongation and initiation. It supports the emerging view that ribosome collisions, stalling, and quality control pathways play active roles in regulating initiation rates in cis. The findings are consistent with recent studies in yeast and metazoans showing translation initiation repression following stalling events. However, the mechanistic details of this feedback remain incompletely understood and merit further investigation, particularly in physiological or stress contexts.

In summary, this is a thoughtfully executed and timely study that provides valuable insights into the dynamic regulation of translation and introduces a modeling framework with broad applicability. It will be of interest to a wide audience in molecular biology, systems biology, and quantitative imaging.

https://doi.org/10.7554/eLife.107160.2.sa3

Reviewer #2 (Public review):

Summary:

This manuscript uses single-molecule run-off experiments and TASEP/HMM models to estimate biophysical parameters, i.e., ribosomal initiation and elongation rates. Combining inferred initiation and elongation rates, the authors quantify ribosomal density. TASEP modeling was used to simulate the mechanistic dynamics of ribosomal translation, and the HMM is used to link ribosomal dynamics to microscope intensity measurements. The authors' main conclusions and findings are:

- Ribosomal elongation rates and initiation rates are strongly coordinated.

- Elongation rates were estimated between 1 and 4.5 aa/sec. Initiation rates were estimated between 1 and 2 ribosomes/min. These values agree with previously reported ones.

- Ribosomal density was determined to be below 12% for all constructs and conditions.

- eIF5A-perturbations (GC7 inhibition) resulted in non-significant changes in translational bursting and ribosome density.

- eIF5A perturbations affected both elongation and initiation rates.

Strengths:

This manuscript presents an interesting scientific hypothesis to study ribosome initiation and elongation concurrently. This topic is relevant for the field. The manuscript presents a novel quantitative methodology to estimate ribosomal initiation rates from Harringtonine run-off assays. This is relevant because run-off assays have been used to estimate, exclusively, elongation rates.

Comments on revisions:

The authors have addressed my concerns. Specifically, they have expanded the discussion on unexpected eIF5A perturbation results, calculated CAI values for all constructs, and made code and data publicly available via GitHub and Zenodo. The mathematical notation is now consistent, and all variables are properly defined.

https://doi.org/10.7554/eLife.107160.2.sa2

Reviewer #3 (Public review):

Disclaimer:

My expertise is in live single-molecule imaging of RNA and transcription, as well as associated data analysis and modeling. While this aligns well with the technical aspects of the manuscript, my background in translation is more limited, and I am not best positioned to assess the novelty of the biological conclusions.

Summary:

This study combines live-cell imaging of nascent proteins on single mRNAs with time-series analysis to investigate the kinetics of mRNA translation.

The authors (i) used a calibration method for estimating absolute ribosome counts, and (ii) developed a new Bayesian approach to infer ribosome counts over time from run-off experiments, enabling estimation of elongation rates and ribosome density across conditions.

They report (i) translational bursting at the single-mRNA level, (ii) low ribosome density (~10% occupancy {plus minus} a few percents), (iii) that ribosome density is minimally affected by perturbations of elongation (using a drug and/or different coding sequences in the reporter), suggesting a homeostatic mechanism potentially involving a feedback of elongation onto initiation, although (iv) this coupling breaks down upon knockout of elongation factor eIF5A.

Strengths:

(1) The manuscript is well written and the conclusions are in general appropriately cautious (besides the few improvements I suggest below).

(2) The time-series inference method is interesting and promising for broader application.

(3) Simulations provide convincing support for the modeling (though some improvements are possible).

(4) The reported homeostatic effect on ribosome density is surprising and carefully validated with multiple perturbations.

(5) Imaging quality and corrections (e.g., flat-fielding, laser power measurements) are robust.

(6) Mathematical modeling is clearly described and precise; a few clarifications could improve it further.

Weaknesses:

(1) The absolute quantification of ribosome numbers (via the measurement of i_{MP}) should be improved. This only affects the finding that ribosome density is low, not that it appears to be under homeostatic control. However, if i_{MP} turns out to be substantially overestimated (hence ribosome density underestimated), then "ribosomes queuing up to the initiation site and physically blocking initiation" could become a relevant hypothesis. In my first review of this work, I made recommendations, which the authors did not follow. In my view, the robustness of this particular aspect of this study remains moderate.

(2) The proposed initiation-elongation coupling is plausible, but alternative explanations such as changes in abortive elongation frequency should be considered. In their response to my previous comments, the authors indicate that this is "beyond the scope of the present work".

(3) More an opportunity for improvement than a weakness: It is unclear what the single-mRNA nature of the inference method is bringing since it is only used here to report _average_ ribosome elongation rate and density (averaged across mRNAs and across time during the run-off experiments -although the method, in principle, has the power to resolve these two aspects). In response to my previous comment, the authors note that such analyses could be incorporated in future work.

https://doi.org/10.7554/eLife.107160.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *In this study, Lamberti et al. investigate how translation initiation and elongation are coordinated at the single-mRNA level in mammalian cells. The authors aim to uncover whether and how cells dynamically adjust initiation rates in response to elongation dynamics, with the overarching goal of understanding how translational homeostasis is maintained. To this end, the study combines single-molecule live-cell imaging using the SunTag system with a kinetic modeling framework grounded in the Totally Asymmetric Simple Exclusion Process (TASEP). By applying this approach to custom reporter constructs with different coding sequences, and under perturbations of the*

> *initiation/elongation factor eIF5A, the authors infer initiation and elongation rates from individual mRNAs and examine how these rates covary.*
>
> *The central finding is that initiation and elongation rates are strongly correlated across a range of coding sequences, resulting in consistently low ribosome density ({less than or equal to}12% of the coding sequence occupied). This coupling is preserved under partial pharmacological inhibition of eIF5A, which slows elongation but is matched by a proportional decrease in initiation, thereby maintaining ribosome density. However, a complete genetic knockout of eIF5A disrupts this coordination, leading to reduced ribosome density, potentially due to changes in ribosome stalling resolution or degradation.*
>
> *Strengths:*
>
> *A key strength of this work is its methodological innovation. The authors develop and validate a TASEP-based Hidden Markov Model (HMM) to infer translation kinetics at single-mRNA resolution. This approach provides a substantial advance over previous population-level or averaged models and enables dynamic reconstruction of ribosome behavior from experimental traces. The model is carefully benchmarked against simulated data and appropriately applied. The experimental design is also strong. The authors construct matched SunTag reporters differing only in codon composition in a defined region of the coding sequence, allowing them to isolate the effects of elongation-related features while controlling for other regulatory elements. The use of both pharmacological and genetic perturbations of eIF5A adds robustness and depth to the biological conclusions. The results are compelling: across all constructs and conditions, ribosome density remains low, and initiation and elongation appear tightly coordinated, suggesting an intrinsic feedback mechanism in translational regulation. These findings challenge the classical view of translation initiation as the sole rate-limiting step and provide new insights into how cells may dynamically maintain translation efficiency and avoid ribosome collisions.*

We thank the reviewer for their constructive assessment of our work, and for recognizing the methodological innovation and experimental rigor of our study.

> *Weaknesses:*
>
> *A limitation of the study is its reliance on exogenous reporter mRNAs in HeLa cells, which may not fully capture the complexity of endogenous translation regulation. While the authors acknowledge this, it remains unclear how generalizable the observed coupling is to native mRNAs or in different cellular contexts.*

We agree that the use of exogenous reporters is a limitation inherent to the SunTag system, for which there is currently no simple alternative for single-mRNA translation imaging. However, we believe our findings are likely generalizable for several reasons.

As discussed in our introduction and discussion, there is growing mechanistic evidence in the literature for coupling between elongation (ribosome collisions) and initiation via pathways such as the GIGYF2-4EHP axis (Amaya et al. 2018, Hickey et al. 2020, Juszkiewicz et al. 2020), which might operate on both exogenous and endogenous mRNAs.

As already acknowledged in our limitations section, our exogenous reporters may not fully recapitulate certain aspects of endogenous translation (e.g., ER-coupled collagen processing), yet the observed initiation-elongation coupling was robust across all tested constructs and conditions.

We have now expanded the Discussion (L393-395) to cite complementary evidence from Dufourt et al. (2021), who used a CRISPR-based approach in Drosophila embryos to measure translation of endogenous genes. We also added a reference to Choi et al. 2025, who uses a ER-specific SunTag reporter to visualize translation at the ER (L395-397).

> *Additionally, the model assumes homogeneous elongation rates and does not explicitly account for ribosome pausing or collisions, which could affect inference accuracy, particularly in constructs designed to induce stalling. While the model is validated under low-density assumptions, more work may be needed to understand how deviations from these assumptions affect parameter estimates in real data.*

We agree with the reviewer that the assumption of homogeneous elongation rates is a simplification, and that our work represents a first step towards rigorous single-trace analysis of translation dynamics. We have explicitly tested the robustness of our model to violations of the low-density assumption through simulations (Figure 2 - figure supplement 2). These show that while parameter inference remains accurate at low ribosome densities, accuracy slightly deteriorates at higher densities, as expected. In fact, our experimental data do provide evidence for heterogeneous elongation: the waiting times between termination events deviate significantly from an exponential distribution (Figure 3 - figure supplement 2C), indicating the presence of ribosome stalling and/or bursting, consistent with the reviewer's concern. We acknowledge in the Limitations section (L402-406) that extending the model to explicitly capture transcript-dependent elongation rates and ribosome interactions remains challenging. The TASEP is difficult to solve analytically under these conditions, but we note that simulation-based inference approaches, such as particle filters to replace HMMs, could provide a path forward for future work to capture this complexity at the single-trace level.

> *Furthermore, although the study observes translation "bursting" behavior, this is not explicitly modeled. Given the growing recognition of translational bursting as a regulatory feature, incorporating or quantifying this behavior more rigorously could strengthen the work's impact.*

While we do not explicitly model the bursting dynamics in the HMM framework, we have quantified bursting behavior directly from the data. Specifically, we measure the duration of translated (ON) and untranslated (OFF) periods across all reporters and conditions (Figure 1G for control conditions and Figure 4G-H for perturbed conditions), finding that active translation typically lasts 10-15 minutes interspersed with shorter silent periods of 5-10 minutes. This empirical characterization demonstrates that bursting is a consistent feature of translation across our experimental conditions. The average duration of silent periods is similar to what was inferred by Livingston et al. 2023 for a similar SunTag reporter; while the average duration of active periods is substantially shorter (~15 min instead of ~40 min), which is consistent with the shorter trace duration in our system compared to theirs (~15 min compared to ~80 min, on average). Incorporating an explicit two-state or multi-state bursting model into the TASEP-HMM framework would indeed be computationally intensive and represents an important direction for future work, as it would enable inference of switching rates alongside initiation and elongation parameters. We have added this point to the Discussion (L415-417).

> *Assessment of Goals and Conclusions:*
>
> *The authors successfully achieve their stated aims: they quantify translation initiation and elongation at the single-mRNA level and show that these processes are dynamically coupled to maintain low ribosome density. The modeling framework is well suited to this task, and the conclusions are supported by multiple lines of evidence, including inferred kinetic parameters, independent ribosome counts, and consistent behavior under perturbation.*
>
> *Impact and Utility:*

This work makes a significant conceptual and technical contribution to the field of translation biology. The modeling framework developed here opens the door to more detailed and quantitative studies of ribosome dynamics on single mRNAs and could be adapted to other imaging systems or perturbations. The discovery of initiation-elongation coupling as a general feature of translation in mammalian cells will likely influence how researchers think about translational regulation under homeostatic and stress conditions.

The data, models, and tools developed in this study will be of broad utility to the community, particularly for researchers studying translation dynamics, ribosome behavior, or the effects of codon usage and mRNA structure on protein synthesis.

Context and Interpretation:

This study contributes to a growing body of evidence that translation is not merely controlled at initiation but involves feedback between elongation and initiation. It supports the emerging view that ribosome collisions, stalling, and quality control pathways play active roles in regulating initiation rates in cis. The findings are consistent with recent studies in yeast and metazoans showing translation initiation repression following stalling events. However, the mechanistic details of this feedback remain incompletely understood and merit further investigation, particularly in physiological or stress contexts.

In summary, this is a thoughtfully executed and timely study that provides valuable insights into the dynamic regulation of translation and introduces a modeling framework with broad applicability. It will be of interest to a wide audience in molecular biology, systems biology, and quantitative imaging.

We appreciate the reviewer's thorough and positive assessment of our work, and that they recognize both the technical innovation of our modeling framework and its potential broad utility to the translation biology community. We agree that further mechanistic investigation of initiation-elongation feedback under various physiological contexts represents an important direction for future research.

Reviewer #2 (Public review):

Summary:

This manuscript uses single-molecule run-off experiments and TASEP/HMM models to estimate biophysical parameters, i.e., ribosomal initiation and elongation rates. Combining inferred initiation and elongation rates, the authors quantify ribosomal density. TASEP modeling was used to simulate the mechanistic dynamics of ribosomal translation, and the HMM is used to link ribosomal dynamics to microscope intensity measurements. The authors' main conclusions and findings are:

(1) Ribosomal elongation rates and initiation rates are strongly coordinated.

(2) Elongation rates were estimated between 1-4.5 aa/sec. Initiation rates were estimated between 0.5-2.5 events/min. These values agree with previously reported values.

(3) Ribosomal density was determined below 12% for all constructs and conditions.

(4) eIF5A-perturbations (KO and GC7 inhibition) resulted in non-significant changes in translational bursting and ribosome density.

(5) eIF5A perturbations resulted in increases in elongation and decreases in initiation rates.

> *Strengths:*
>
> *This manuscript presents an interesting scientific hypothesis to study ribosome initiation and elongation concurrently. This topic is highly relevant for the field. The manuscript presents a novel quantitative methodology to estimate ribosomal initiation rates from Harringtonine run-off assays. This is relevant because run-off assays have been used to estimate, exclusively, elongation rates.*

We thank the reviewer for their careful evaluation of our work and for recognizing the novelty of our quantitative methodology to extract both initiation and elongation rates from harringtonine run-off assays, extending beyond the traditional use of these experiments.

> *Weaknesses:*
>
> *The conclusion of the strong coordination between initiation and elongation rates is interesting, but some results are unexpected, and further experimental validation is needed to ensure this coordination is valid.*

We agree that some of our findings need further experimental investigation in future studies. However, we believe that the coordination between initiation and elongation is supported by multiple results in our current work: (1) the strong correlation observed across all reporters and conditions (Figure 3E), and (2) the consistent maintenance of low ribosome density despite varying elongation rates. While additional experimental validation would be valuable, we note that directly manipulating initiation or elongation independently in mammalian cells remains technically challenging. Nevertheless, our findings are consistent with emerging mechanistic understanding of collision-sensing pathways (GIGYF2-4EHP) that could mediate such coupling, as discussed in our manuscript.

> *(1) eIF5a perturbations resulted in a non-significant effect on the fraction of translating mRNA, translation duration, and bursting periods. Given the central role of eIF5a, I would have expected a different outcome. I would recommend that the authors expand the discussion and review more literature to justify these findings.*

We appreciate this comment. This finding is indeed discussed in detail in our manuscript (Discussion, paragraphs 6-7). As we note there, while eIF5A plays a critical role in elongation, the maintenance of bursting dynamics and ribosome density upon perturbation can be explained by compensatory feedback mechanisms. Specifically, the coordinated decrease in initiation rates that counterbalances slower elongation to maintain homeostatic ribosome density. We also discuss several factors that complicate interpretation: (1) potential RQC-mediated degradation masking stronger effects in proline-rich constructs, (2) differences between GC7 treatment and genetic knockout suggesting altered stalling resolution kinetics, and (3) the limitations of using exogenous reporters that lack ER-coupled processing, which may be critical for eIF5A function in endogenous collagen translation (as suggested by Rossi et al., 2014; Mandal et al., 2016; Barba-Aliaga et al., 2021). The mechanistic complexity and tissue-specific nature of eIF5A function in mammals, which differs substantially from the better-characterized yeast system, likely contributes to the nuanced phenotype we observe. We believe our Discussion adequately addresses these points.

> *(2) The AAG construct leading to slow elongation is very surprising. It is the opposite of the field consensus, where codon-optimized gene sequences are expected to elongate faster. More information about each construct should be provided. I would recommend more bioinformatic analysis on this, for example, calculating CAI for all constructs, or predicting the structures of the proteins.*

We agree that the slow elongation of the AAG construct is counterintuitive and indeed surprising. Following the reviewer's suggestion, we have now calculated the Codon

Adaptation Index (CAI) for all constructs (Renilla 0.89, Col1a1 0.78, Col1a1 mutated 0.74). It is therefore unlikely that codon bias explains the slow translation, particularly since we designed the mutated Col1a1 construct with alanine codons selected to respect human codon usage bias, thereby minimizing changes in codon optimality. As we discuss in the manuscript, we hypothesize that the proline-to-alanine substitutions disrupted co-translational folding of the collagen-derived sequence. Prolines are critical for collagen triple-helix formation (Shoulders and Raines, 2009), and their replacement with alanines likely generates misfolded intermediates that cause ribosome stalling (Barba-Aliaga et al., 2021; Komar et al., 2024). This interpretation is supported by the high frequency (>30%) of incomplete run-off traces for AAG, suggesting persistent stalling events. Our findings thus illustrate an important potential caveat: "optimizing" a sequence based solely on codon usage can be detrimental when it disrupts functionally important structural features or co-translational folding pathways.

This highlights that elongation rates depend not only on codon optimality but also on the interplay between nascent chain properties and ribosome progression.

> *(3) The authors should consider using their methodology to study the effects of modifying the 5'UTR, resulting in changes in initiation rate and bursting, such as previously shown in reference Livingston et al., 2023. This may be outside of the scope of this project, but the authors could add this as a future direction and discuss if this may corroborate their conclusions.*

We thank the reviewer for this excellent suggestion. We agree that applying our methodology to 5'-UTR variants would provide a complementary test of initiation-elongation coupling, and we have now added this as a future direction in the Discussion (L417-420).

> *(4) The mathematical model and parameter inference routines are central to the conclusions of this manuscript. In order to support reproducibility, the computational code should be made available and well-documented, with a requirements file indicating the dependencies and their versions.*

We have added the Github link in the manuscript (*https://github.com/naef-lab/suntag-analysis*) and have also deposited the data (.ome.tif) on Zenodo (*https://zenodo.org/records/17669332*).

> ***Reviewer #3 (Public review):***
>
> *Disclaimer:*
>
> *My expertise is in live single-molecule imaging of RNA and transcription, as well as associated data analysis and modeling. While this aligns well with the technical aspects of the manuscript, my background in translation is more limited, and I am not best positioned to assess the novelty of the biological conclusions.*
>
> *Summary:*
>
> *This study combines live-cell imaging of nascent proteins on single mRNAs with time-series analysis to investigate the kinetics of mRNA translation.*
>
> *The authors (i) used a calibration method for estimating absolute ribosome counts, and (ii) developed a new Bayesian approach to infer ribosome counts over time from run-off experiments, enabling estimation of elongation rates and ribosome density across conditions.*
>
> *They report (i) translational bursting at the single-mRNA level, (ii) low ribosome density (~10% occupancy*

> *{plus minus} a few percents), (iii) that ribosome density is minimally affected by perturbations of elongation (using a drug and/or different coding sequences in the reporter), suggesting a homeostatic mechanism potentially involving a feedback of elongation onto initiation, although (iv) this coupling breaks down upon knockout of elongation factor eIF5A.*
>
> *Strengths:*
>
> *(1) The manuscript is well written, and the conclusions are, in general, appropriately cautious (besides the few improvements I suggest below).*
>
> *(2) The time-series inference method is interesting and promising for broader applications.*
>
> *(3) Simulations provide convincing support for the modeling (though some improvements are possible).*
>
> *(4) The reported homeostatic effect on ribosome density is surprising and carefully validated with multiple perturbations.*
>
> *(5) Imaging quality and corrections (e.g., flat-fielding, laser power measurements) are robust.*
>
> *(6) Mathematical modeling is clearly described and precise; a few clarifications could improve it further.*

We thank the reviewer for recognizing the novelty of the approach and its rigour, and for providing suggestions to improve it further.

> *Weaknesses:*
>
> *(1) The absolute quantification of ribosome numbers (via the measurement of i_{MP}) should be improved.This only affects the finding that ribosome density is low, not that it appears to be under homeostatic control. However, if i_{MP} turns out to be substantially overestimated (hence ribosome density underestimated), then "ribosomes queuing up to the initiation site and physically blocking initiation" could become a relevant hypothesis. In my detailed recommendations to the authors, I list points that need clarification in their quantifications and suggest an independent validation experiment (measuring the intensity of an object with a known number of GFP molecules, e.g., MS2-GFP MS2-GFP-labeled RNAs, or individual GEMs).*

We agree with the reviewer that the estimation of the number of ribosomes is central to our finding that translation happens at low density on our reporters. This result derives from our measurement of the intensity of one mature protein (i_{MP}), that we have achieved by using a SunTag reporter with a RH1 domain in the C terminus of the mature protein, allowing us to stabilise mature proteins via actin-tethering. In addition, as suggested by the reviewer, we already validated this result with an independent estimate of the mature protein intensity (Figure 5 - figure supplement 2B), which was obtained by adding the mature protein intensity directly as a free parameter of the HMM. The inferred value of mature protein intensity for each construct (10-15 a.u) was remarkably close to the experimental calibration result (14 ± 2 a.u.). Therefore, we have confidence that our absolute quantification of ribosome numbers is accurate.

> *(2) The proposed initiation-elongation coupling is plausible, but alternative explanations, such as changes in abortive elongation frequency, should be considered more carefully. The authors mention this possibility, but should test or rule it out quantitatively.*

We thank the reviewer for the comment, but we consider that ruling out alternative explanations through new perturbation experiments is beyond the scope of the present work.

> *(3) The observation of translational bursting is presented as novel, but similar findings were reported by Livingston et al. (2023) using a similar SunTag-MS2 system. This prior work should be acknowledged, and the added value of the current approach clarified.*

We did cite Livingston et al. (2023) in several places, but we recognized that we could add a few citations in key places, to make clear that the observation of bursting is not novel but is in agreement with previous results. We now did so in the Results and Discussion sections.

> *(4) It is unclear what the single-mRNA nature of the inference method is bringing since it is only used here to report _average_ ribosome elongation rate and density (averaged across mRNAs and across time during the run-off experiments - although the method, in principle, has the power to resolve these two aspects).*

While decoding individual traces, our model infers shared (population-level) rates. Inferring transcript-specific parameters would be more informative, but it is highly challenging due to the uncertainty on the initial ribosome distribution on single transcripts. Pooling multiple transcripts together allows us to use some assumptions on the initial distribution and infer average elongation and initiation-rate parameters, while revealing substantial mRNA-to-mRNA variability in the posterior decoding (e.g. Figure 3 - figure Supplement 2C). Indeed, the inference still informs on the single-trace run-off time distribution (Figure 3 A) and the waiting time between termination events (Figure 3 - figure supplement 2C), suggesting the presence of stalling and bursting. In addition, the transcript-to-transcript heterogeneity is likely accounted for by our model better than previous methods (linear fit of the average run-off intensity), as suggested by their comparison (Figure 3 - figure supplement 2 A). In the future the model could be refined by introducing transcript-specific parameters, possibly in a hierarchical way, alongside shared parameters.

> *(5) I did not find any statement about data availability. The data should be made available. Their absence limits the ability to fully assess and reproduce the findings.*

We have added the Github link in the manuscript (*https://github.com/naef-lab/suntag-analysis*) and have also deposited the data (.ome.tif) on Zenodo (*https://zenodo.org/records/17669332*).

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *Major Comments:*
>
> *(1) Lack of Explicit Bursting Model*
>
> *Although translation "bursts" are observed, the current framework does not explicitly model initiation as a stochastic ON/OFF process. This limits insight into regulatory mechanisms controlling burst frequency or duration. The authors should either incorporate a two-state/more-state (bursting) model of initiation or perform statistical analysis (e.g., dwell-time distributions) to quantify bursting dynamics. They should clarify how bursting influences the interpretation of initiation rate estimates.*

We agree with the reviewer that an explicit bursting model (e.g., a two-state telegraph model) would be the ideal theoretical framework. However, integrating such a model into the TASEP-HMM inference framework is computationally intensive and complex. As a robust first step, we have opted to quantify bursting empirically based on the decoded single-mRNA traces. As shown in Figure 1G (control) and Figure 4G (perturbed conditions), we explicitly measured the duration of "ON" (translated) and "OFF" (untranslated) periods. This statistical analysis provides a quantitative description of the bursting dynamics without relying on the specific assumptions of a telegraph model. We have clarified this in the text (L123-125) and, as suggested, added a discussion (L415-417) on the potential extensions of the model to include explicit switching kinetics in the Outlook section.

> *(2) Assumption of Uniform Elongation Rates*
>
> *The model assumes homogeneous elongation across coding sequences, which may not hold for stalling-prone inserts (e.g., PPG). This simplification could bias inference, particularly in cases of sequence-specific pausing. Adding simulations or sensitivity analysis to assess how non-uniform elongation affects the accuracy of inferred parameters. The authors should explicitly discuss how ribosome stalling, collisions, or heterogeneity might skew model outputs (see point 4).*

A strong stalling sequence that affects all ribosomes equally should not deteriorate the inference of the initiation rate, provided that the low-density assumption holds. The scenario where stalling events lead to higher density, and thus increased ribosome-ribosome interactions, is comparable to the conditions explored in Figure 2E. In those simulations, we tested the inference on data generated with varying initiation and elongation rates, resulting in ribosome densities ranging from low to high. We demonstrated that the inference remains robust at low ribosome densities (<10%). At higher densities, the accuracy of the initiation rate estimate decreases, whereas the elongation rate estimate remains comparatively robust. Additionally, the model tends to overestimate ribosome density under high-density conditions, likely because it neglects ribosome interference at the initiation site (Figure 2 figure supplement 2C). We agree that a deeper investigation into the consequences of stochastic stalling and bursting would be beneficial, and we have explicitly acknowledged this in the Limitations section.

> *(3) Interpretation of eIF5A Knockout Phenotype*
>
> *The observation that eIF5A KO reduces initiation more than elongation, leading to decreased ribosome density, is biologically intriguing. However, the explanation invoking altered RQC kinetics is speculative and not directly tested. The authors should consider validating the RQC hypothesis by monitoring reporter mRNA stability, ribosome collision markers, or translation termination intermediates.*

We thank the reviewer for the comment, but we consider that ruling out alternative explanations through new experiments is beyond the scope of the present work.

> *(4) To strengthen the manuscript, the authors should incorporate insights from three studies.*
>
> *- Livingston et al. (PMC10330622) found that translation occurs in bursts, influenced by mRNA features and initiation factors, supporting the coupling of initiation and elongation.*
>
> *- Madern et al. (PMID: 39892379) demonstrated that ribosome cooperativity enhances translational efficiency, highlighting coordinated ribosome behavior.*
>
> *- Dufourt et al. (PMID: 33927056) observed that high initiation rates correlate with high elongation rates, suggesting a conserved mechanism across cell cultures and organisms.*
>
> *Integrating these studies could enrich the manuscript's interpretation and stimulate new avenues of thought.*

We thank the reviewer for the valuable comment. We added citations of Livingston et al. in the context of translational bursting. We already cited Madern et al. in multiple places and, although its observations of ribosome cooperativity are very compelling, they cannot be linked with our observations of a feedback between initiation and elongation, and it would be very challenging to see a similar effect on our reporters. This is why we did not expressly discuss cooperativity. We also integrated Dufourt et al. in the Discussion about the possibility of designing genetically-encoded reporter. We also added a sentence about the possibility of using an ER-specific SunTag reporter, as done recently in Choi et al., Nature (2025) (*https://doi.org/10.1038/s41586-025-09718-0*).

> *Minor Comments:*
>
> *(1) Use consistent naming for SunTag reporters (e.g., "PPG" vs "proline-rich") throughout.*

Thank you for the comment. However, the term proline-rich always appears together with PPG, so we believe that the naming is clear and consistent.

> *(2) Consider a schematic overview of the experimental design and modeling pipeline for accessibility.*

Thank you for the suggestion. We consider that experimental design and modeling is now sufficiently clearly described and does not justify an additional scheme.

> *(3) Clarify how incomplete run-off traces are handled in the HMM inference.*

Incomplete run-off traces are treated identically to complete traces in our HMM inference. This is possible because our model relies on the probability of transitions occurring per time step to infer rates. It does not require observing the final "empty" state to estimate the kinetic parameters α and λ. The loss of signal (e.g., mRNA moving out of the focal volume or photobleaching) does not invalidate the kinetic information contained in the portion of the trace that was observed. We have clarified this in the Methods section.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) Reproducibility:*
>
> *(1.1) The authors should use a GitHub repository with a timestamp for the release version.*

The code is available on GitHub (*https://github.com/naef-lab/suntag-analysis*).

> *(1.2) Make raw images and data available in a figure repository like Figshare.*

The raw images (.ome.tif) are now available on Zenodo (*https://zenodo.org/records/17669332*).

> *(2) Paper reorganization and expansion of the intensity and ribosome quantification:*
>
> *(2.1) Given the relevance of the initiation and elongation rates for the conclusions of this study, and the fact that the authors inferred these rates from the spot intensities. I recommend that the authors move Figure 1 Supplement 2 to the main text and expand the description of the process to relate spot intensity and number of ribosomes. Please also expand the figure caption for this image.*

We agree with the importance of this validation. We have expanded the description of the calibration experiment in the main text and in the figure caption.

> *(2.2) I suggest the authors explicitly mention the use of HMM in the abstract.*

We have now explicitly mentioned the TASEP-based HMM in the abstract.

> *(2.3) In line 492, please add the frame rate used to acquire the images for the run-off assays.*

We have added the specific frame rate (one frame every 20 seconds) to the relevant section.

> *(3) Figures and captions:*
>
> *(3.1) Figure 1, Supplement 2. Please add a description of the colors used in plots B, C.*

We have expanded the caption and added the color description.

> *(3.2) In the Figure 2 caption. It is not clear what the authors mean by "traceseLife". Please ensure it is not a typo.*

Thank you for spotting this. We have corrected the typo.

> *(3.3) Figure 1 A, in the cartoon N(alpha)->N-1, shouldn't the transition also depend on lambda?*

The transition probability was explicitly derived in the "Bayesian modeling of run-off traces" section (Eqs. 17-18), and does not depend on λ, but only on the initiation rate under the low-density assumption.

> *(3.4) Figure 3, Supplement 2. "presence of bursting and stalling.." has a typo.*

Corrected.

> *(3.5) Figure 5, panel C, the y-axis label should be "run-off time (min)."*

Corrected.

> *(3.6) For most figures, add significance bars.*
>
> *(3.7) In the figure captions, please add the total number of cells used for each condition.*

We have systematically indicated the number of traces (n_t) and the number of independent experiments (n_e) in the captions in this format (n_t, n_e).

> *(4) Mathematical Methods:*

We greatly thank the reviewer for their detailed attention to the mathematical notation. We have addressed all points below.

> *(4.1) In lines 555, Materials and Methods, subsection, Quantification of Intensity Traces, multiple equations are not numbered. For example, after Equation (4), no numbers are provided for the rest of the equations. Please keep consistency throughout the whole document.*

We have ensured that all equations are now consistently numbered throughout the document.

> *(4.2) In line 588, the authors mention "X is a standard normal random variable with mean μ and standard deviation s_0". Please ensure this is correct. A standard normal random variable has a 0 mean and std 1.*

Thank you for the suggestion, we have corrected the text (L678).

> *(4.3) Line 546, Equation 2. The authors use mu(x,y) to describe a 2d Gaussian function. But later in line 587, the authors reuse the same variable name in equation 5 to redefine the intensity as mu = b_0 + I.*

We have renamed the 2D Gaussian function to \mu_{2D}(x,y) in the spot tracking section

> *(4.4) For the complete document, it could be beneficial to the reader if the authors expand the definition of the relationship between the signal "y" and the spot intensity "I". Please note how the paragraph in lines 582-587 does not properly introduce "y".*

We have added an explicit definition of y and its relationship to the underlying spot intensity I in the text to improve readability and clarity.

> *(4.5) Please ensure consistency in variable names. For example, "I" is used in line 587 for the experimental spot intensity, then line 763 redefines I(t) as the total intensity obtained from the TASEP model; please use "I_sim(t)" for simulated intensities. Please note that reusing the variable "I" for different contexts makes it hard for the reader to follow the text.*

We agree that this was confusing. We have implemented the suggestion and now distinguish simulated intensities using the notation I_S .

> *(4.6) Line 555 "The prior on the total intensity I is an "uninformative" prior" I ~ half_normal(1000). Please ensure it is not "I_0 ~ half_normal(1000)."?*

We confirm that "I" is the correct variable representing the total intensity in this context; we do not use an "I_0" variable here.

> *(4.7) In lines 595, equation 6. Ensure that the equation is correct. Shouldn't it be: s_0^2 = ln (1 + (sigma_meas^2 / ⟨y⟩^2))? Please ensure that this is correct and it is not affecting the calculated values given in lines 598.*

Thank you for catching this typo. We have corrected the equation in the manuscript. We confirm that the calculations performed in the code used the correct formula, so the reported values remain unchanged.

> *(4.8) In line 597, "the mean intensity square ^2". Please ensure it is not "the square of the temporal mean intensity."*

We have corrected the text to "the square of the temporal mean intensity."

> *(4.9) In lines 602-619, Bayesian modeling of run-off traces, please ensure to introduce the constant "\ell". Used to define the ribosomal footprint?*

We have added the explicit definition of ℓ as the ribosome footprint size (length of transcript occupied by one ribosome) in the "Bayesian modeling of run-off traces" section.

> *(4.10) Line 687 has a minor typo "[...] ribosome distribution.. Then, [...]"*

We have corrected the punctuation.

> *(4.11) In line 678, Equation 19 introduces the constant "L_S", Please ensure that it is defined in the text.*

We have added the explicit definition of L_S (the length of the SunTag) to the text surrounding Equation 19.

> *(4.12) In line 695, Equation 22, please consider using a subscript to differentiate the variance due to ribosome configuration. For example, instead of "sigma (...)^2" use something like "sigma_c ^2 (...)". Ensure that this change is correctly applied to Equation 24 and all other affected equations.*

Thank you, we have implemented the suggestions.

> *(4.13) In line 696, please double-check equations 26 and 27. Specifically, the denominator ^2. Given the previous text, it is hard to follow the meaning of this variable.*

We have revised the notation in Equations 26 and 27 to ensure the denominator is consistent with the definitions provided in the text.

> *(4.14) In lines 726, the authors mention "[...], but for the purposes of this dissertation [...]", it should be "[...], but for the purposes of this study [...]"*

Thank you for spotting this. We have replaced "dissertation" with "study."

> *(4.15) Equations 5, 28, 37, and the unnumbered equation between Equations 16 and 17 are similar, but in some, "y" does not explicitly depend on time. Please ensure this is correct.*

We have verified these equations and believe they are correct.

> *(4.16) Please review the complete document and ensure that variables and constants used in the equations are defined in the text. Please ensure that the same variable names are not reused for different concepts. To improve readability and flow in the text, please review the complete Materials and Methods sections and evaluate if the modeling section can be written more clearly and concisely. For example, Equation 28 is repeated in the text.*

We have performed a comprehensive review of the Materials and Methods section. To improve conciseness and flow, we have merged the subsection “Observation model and estimation of observation parameters” with the “Bayesian modeling of run-off traces” section. This allowed us to remove redundant definitions and repeated equations (such as the previous Equation 28). We have also checked that all variables and constants are defined upon first use and that variable names remain consistent throughout the manuscript.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) Data Presentation*
>
> *(1.1) In main Figures 1D and 4E, the traces appear to show frequent on-off-on transitions ("bursting"), but in supplementary figures (1-S1A and 4-S1A), this behavior is seen in only ~8 of 54 traces. Are the main figure examples truly representative?*

We acknowledge the reviewer's point. In Figure 1D, we selected some of the longest and most illustrative traces to highlight the bursting dynamics. We agree that the term "representative" might be misleading if interpreted as "average." We have updated the text to state "we show bursting traces" to more accurately reflect the selection.

> *(1.2) There are 8 videos, but I could not identify which is which.*

Thank you for pointing this out. We have renamed the video files to clearly correspond to the figures and conditions they represent.

> *(2) Data Availability:*
>
> *As noted above, the data should be shared. This is in accordance with eLife's policy: "Authors must make all original data used to support the claims of the paper, or that are required to reproduce them, available in the manuscript text, tables, figures or supplementary materials, or at a trusted digital repository (the latter is recommended). [...] eLife considers works to be published when they are posted as preprints, and expects preprints we review to meet the standards outlined here." Access to the time traces would have been helpful for reviewers.*

We have now added the Github link for the code (*https://github.com/naef-lab/suntag-analysis*) and deposited the raw data (.ome.tif files) on Zenodo (10.5281/zenodo.17669332).

> *(3) Model Assumptions:*
>
> *(3.1) The broad range of run-off times (Figure 3A) suggests stalling, which may be incompatible with the 'low-density' assumption used on the TASEP model, which essentially assumes that ribosomes do not bump into each other. This could impact the validity of the assumptions that ribosomes behave independently, elongate at constant speed (necessary for the continuum-limit approximation), and that the rate-limiting step is the initiation. How robust are the inferences to this assumption?*

We agree that the deviation of waiting times from an exponential distribution (Figure 3 - figure supplement 2C) suggests the presence of stalling, which challenges the strict low-density assumption and constant elongation speed. We explicitly explored the robustness of our model to higher ribosome densities in simulations. As shown in Figure 2 - figure supplement 2, while the model accuracy for single parameters deteriorates at very high densities (overestimating density due to neglected interference), it remains robust for estimating global rates in the regime relevant to our data. We have expanded the discussion on the limitations of the low density and homogeneous elongation rate assumptions in the text (L404-408).

> *(3.2) Since all constructs share the same SunTag region, elongation rates should be identical there and diverge only in the variable region. This would affect $\gamma (t)$ and hence possibly affect the results. A brief discussion would be helpful.*

This is a valid point. Currently, our model infers a single average elongation rate that effectively averages the behavior over the SunTag and the variable CDS regions. Modeling distinct rates for these regions would be a valuable extension but adds significant complexity. While our current "effective rate" approach might underestimate the magnitude of differences between reporters, it captures the global kinetic trend. We have added a brief discussion acknowledging this simplification (L408-412).

> *(3.3) A similar point applies to the Gillespie simulations: modeling the SunTag region with a shared elongation rate would be more accurate.*

We agree. Simulating distinct rates for the SunTag and CDS would increase realism, though our current homogeneous simulations serve primarily to benchmark the inference framework itself. We have noted this as a potential future improvement (L413-414).

> *(3.4) Equation (13) assumes that switching between bursting and non-bursting states is much slower than the elongation time. First, this should be made explicit. Second, this is not quite true (~5 min elongation time on Figure 3-s2A vs ~5-15min switching times on Figure 1). It would be useful to show the intensity distribution at t=0 and compare it to the expected mixture distribution (i.e., a Poisson distribution + some extra 'N=0' cells).*

We thank the reviewer for this insightful comment. We have added a sentence to the text explicitly stating the assumption that switching dynamics are slower than the translation time. While the timescales are indeed closer than ideal (5 min vs. 5-15 min), this assumption allows for a tractable approximation of the initial conditions for the run-off inference. Comparing the intensity distribution at t=0 to a zero-inflated Poisson distribution is an excellent suggestion for validation, which we will consider for future iterations of the model.

> *(4) Microscopy Quantifications:*
>
> *(4.1) Figure 1-S2A shows variable scFv-GFP expression across cells. Were cells selected for uniform expression in the analysis? Or is the SunTag assumed saturated? which would*

> *then need to be demonstrated.*

All cell lines used are monoclonal, and cells were selected via FACS for consistent average cytoplasmic GFP signal. We assume the SunTag is saturated based on the established characterization of the system by Tanenbaum et al. (2014), where the high affinity of the scFv-GFP ensures saturation at expression levels similar to ours.

> *(4.2) As translation proceeds, free scFv-GFP may become limiting due to the accumulation of mature SunTag-containing proteins. This would be difficult to detect (since mature proteins stay in the cytoplasm) and could affect intensity measurements (newly synthesized SunTag proteins getting dimmer over time).*

This effect can occur with very long induction times. To mitigate this, we optimized the Doxycycline (Dox) incubation time for our harringtonine experiments to prevent excessive accumulation of mature protein. We also monitor the cytoplasmic background for granularity, which would indicate aggregation or accumulation.

> *(4.3) The statements "for some traces, the mRNA signal was lost before the run-off completion" (line 195) and "we observed relatively consistent fractions of translated transcripts and trace duration distributions across reporters" (line 340) should be supported by a supplementary figure.*

The first statement is supported by Figure 2 - figure supplement 1, which shows representative run-off traces for all constructs, including incomplete ones.

The second statement regarding consistency is supported by the quantitative data in Figure 1E and G, which summarize the fraction of translated transcripts and trace durations across conditions.

> *(4.4) Measurements of single mature protein intensity i_{MP}:*
>
> *(4.4.1) Since puromycin is used to disassemble elongating ribosomes, calibration may be biased by incomplete translation products (likely a substantial fraction, since the Dox induction is only 20min and RNAs need several minutes to be transcribed, exported, and then fully translated).*

As mentioned in the “Live-cell imaging” paragraph, the imaging takes place 40 min after the end of Dox incubation. This provides ample time for mRNA export and full translation of the synthesized proteins. Consequently, the fraction of incomplete products generated by the final puromycin addition is negligible compared to the pool of fully synthesized mature proteins accumulated during the preceding hour.

> *(4.4.2) Line 519: "The intensity of each spot is averaged over the 100 frames". Do I understand correctly that you are looking at immobile proteins? What immobilizes these proteins? Are these small aggregates? It would be surprising that these aggregates have really only 1, 2, or 3 proteins, as suggested by Figure 1-S2A.*

We are visualizing mature proteins that are specifically tethered to the actin cytoskeleton. This is achieved using a reporter where the RH1 domain is fused directly to the C-terminus of the Renilla protein (SunTag-Renilla-RH1). The RH1 domain recruits the endogenous Myosin Va motor, which anchors the protein to actin filaments, rendering it immobile. Since each Myosin Va motor interacts with one RH1 domain (and thus one mature protein), the resulting spots represent individual immobilized proteins rather than aggregates. We have now revised the text and Methods section to make this calibration strategy and the construct design clearer (L130-140).

> *(4.4.3) Estimating the average intensity i_{MP} of single proteins all resides in the seeing discrete modes in the histogram of Figure 1-S2B, which is not very convincing. A*

> *complementary experiment, measuring *on the same microscope* the intensity of an object with a known number of GFP molecules (e.g., MS2-GFP labeled RNAs, or individual GEMs https://doi.org/10.1016/j.cell.2018.05.042 (only requiring a single transfection)) would be reassuring to convince the reader that we are not off by an order of magnitude.*

While a complementary calibration experiment would be valuable, we believe our current estimate is robust because it is independently validated by our model. When we inferred i_{MP} as a free parameter in the HMM (Figure 5 - figure supplement 2B), the resulting value (10-15 a.u.) was remarkably consistent with our experimental calibration (14 ± 2 a.u.). We have clarified this independent validation in the text to strengthen the confidence in our quantification (L264-272).

> *(4.4.4) Further on the histogram in Figure 1-S2B:*
>
> *- The gap between the first two modes is unexpectedly sharp. Can you double-check? It means that we have a completely empty bin between two of the most populated bins.*

We have double-checked the data; the plot is correct, though the sharp gap is likely due to the small sample size (n=29).

> *- I am surprised not to see 3 modes or more, given that panel A shows three levels of intensity (the three colors of the arrows).*

As noted below, brighter foci exist but fall outside the displayed range of the histogram.

> *- It is unclear what the statistical test is and what it is supposed to demonstrate.*

The Student's t-test compares the means of the two identified populations to confirm they are statistically distinct intensity groups.

> *- I count n = 29, not 31. (The sample is small enough that the bars of the histogram show clear discrete heights, proportional to 1, 2, 3, 4, and 5 --adding up all the counts, I get 29). Is there a mistake somewhere? Or are some points falling outside of the displayed x-range?*

You are correct. Two brighter data points fell outside the displayed range. The total number of foci in the histogram is 29. We have corrected the figure caption and the text accordingly.

> *(5) Miscellaneous Points:*
>
> *(5.1) Panel B in Figure 2-s1 appears to be missing.*

The figure contains only one panel.

> *(5.2) In Equation (7), l is not defined (presumably ribosome footprint length?). Instead, J is defined right after eq (7), as if it were used in this equation.*

Thank you for pointing this out, we have corrected it.

> *(5.3) Line 703, did you mean to write something else than "Equation 26" (since equation 26 is defined after)?*

Yes, this was a typo. We have corrected the cross-reference.

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